

PRODUCTION / IMPROVEMENT / RELIABILITY / HANDOFFS

Manufacturing English

Describe the loss. Test the explanation.
Make the handoff clear.

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Eight lessons. Eleven extended conversations.

INSIDE THE BOOK

192 vocabulary entries / 128 reusable phrases
220 extended-dialogue turns / 11 transfer exchanges
Word banks and explained answer keys

Upper-intermediate to advanced (B2-C1). For production, quality, maintenance, supplier, and continuous-improvement teams.

2026 edition

Eight scenarios. Eight skills.

Eight shop-floor and management conversations connecting production facts to clear, accountable next steps.

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Keep the number attached to its meaning.

Manufacturing discussions move quickly between quantities, causes, schedules, and decisions. A missed target may be real even when its cause is unknown. A held lot may contain uninspected material. A named action may still be unfinished. These cases develop the language for keeping those distinctions visible under the everyday pressure to restore output and meet commitments.

State the measure and its basis

Give the unit, time period, numerator, and comparison. Output attainment, availability, and first-pass yield are different measures; a familiar percentage should not be relabeled without the required data.

"We achieved 84% of the output target; that is not an OEE result."

Observe before assigning a cause

A timeline or repeated symptom can guide investigation without proving a root cause. State what changed, what was observed, and which competing explanations remain open.

"The defect followed the supplier change, but the process settings changed too."

Separate material states

Scrapped, awaiting rework, on hold, accepted, and released have different consequences. Use quantities for each state and avoid counting expected recovery as finished good output.

"Sixty units await rework; they are not yet accepted units."

Make the handoff closed-loop

A sent message does not prove it was understood. Identify the open item, evidence, responsible person, and next checkpoint, then confirm receipt and understanding without implying completion.

"Two checks remain open; please confirm their identifiers and the next update."

PROFESSIONAL BOUNDARIES

Original fictional language practice, not equipment-operation, maintenance, engineering, or safety instructions. Figures, organizations, and workplace cases are invented. Actual work must follow applicable law, approved site procedures, trained personnel, and authorized decisions. Classroom safety discussions do not authorize isolation, servicing, or restart.

Eight hundred forty is not the whole story

A Read the situation

Report an output shortfall precisely while separating observed downtime, unconfirmed causes, and different performance measures.

Production lead Mei and operations manager Daniel review yesterday's line output: 840 completed units against a 1,000-unit target. The log records 40 minutes of downtime, but the cause and its contribution to the shortfall are still being checked. The briefing supplies no planned production time, ideal cycle time, or quality breakdown. A dashboard labels 84% as overall equipment effectiveness. Mei and Daniel must correct the metric and give a useful update without claiming the downtime explains every missing unit.

PEOPLE / Mei: Production lead | Daniel: Operations manager

A PRECISE UPDATE CAN STILL BE BRIEF

Start with the result, then the confirmed loss information and the open cause. This lets a busy manager distinguish the size of the gap from the explanation. Avoid filling a short status slot with a confident causal claim simply because the investigation is unfinished.

Briefing checks

Circle one answer in each item. Use the briefing.

1. What was the output shortfall?

A 40 units

B 840 units

C 1,840 units

D 160 units

2. What does 84% represent here?

A Output as a share of the stated target

B A supplied OEE calculation

C A verified availability percentage

D A first-pass yield result

3. What is known about the downtime?

A It was caused by operator error.

B It proves the line had no quality losses.

C Forty minutes were logged; its cause and contribution remain under review.

D It definitely explains all 160 missing units.

[Answers and explanations: page 85.](#)

The vocabulary that does the work

B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

production target

The planned output quantity for a defined period.

IN USE set a production target

actual output

The quantity actually produced in the stated period.

IN USE report actual output

target attainment

Actual output expressed relative to the stated target.

IN USE calculate target attainment

shortfall

The amount by which a result falls below its target.

IN USE quantify the shortfall

throughput

The quantity passing through a process per unit of time.

IN USE measure throughput

cycle time

The elapsed time required for a defined process cycle.

IN USE measure cycle time

takt time

Available production time divided by required demand for the same period.

IN USE calculate takt time

lead time

The elapsed time from a defined request or start to completion.

IN USE reduce lead time

work in process

Material that has entered production but is not yet finished.

IN USE control work in process

bottleneck

The step that limits the output of the overall process.

IN USE identify the bottleneck

constraint

A limiting factor on the system's performance.

IN USE manage the constraint

line balance

Distribution of work across production steps.

IN USE improve line balance

Precision check

840 divided by 1,000 is 84% target attainment. The shortfall is 160 units, or 16% of target. Neither result supplies OEE, availability, or first-pass yield. Those measures require their own defined inputs and calculation bases.

The vocabulary that does the work

B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

downtime

Time during which equipment or a process is not producing as defined.

IN USE log downtime

planned stop

A scheduled interruption included in the operating plan.

IN USE distinguish planned stops

unplanned stop

An interruption not included in the intended operating schedule.

IN USE investigate unplanned stops

availability

The share of planned production time in which equipment is running.

IN USE assess equipment availability

performance loss

Output-speed loss relative to the defined ideal running rate.

IN USE analyze performance losses

quality loss

Output that does not meet the defined good-product criteria.

IN USE quantify quality losses

overall equipment effectiveness

Availability multiplied by performance and quality, with defined factors; abbreviated OEE.

IN USE calculate overall equipment effectiveness

daily management

Routine review of performance, problems, and follow-up actions.

IN USE support daily management

tier meeting

A short escalation meeting at a defined organizational level.

IN USE escalate through a tier meeting

visual board

A shared display of current measures, status, and actions.

IN USE update the visual board

loss attribution

Assignment of a measured loss to supported contributing factors.

IN USE verify loss attribution

recovery plan

A defined proposal for addressing a performance or schedule gap.

IN USE review the recovery plan

Precision check

A recorded 40-minute stop does not automatically explain the entire output gap. The expected running rate, other losses, and the time basis matter. Describe the log as evidence under review rather than a completed causal analysis.

Phrases you can actually use

C Notice the language / 1 of 2

LEAD WITH THE RESULT

"We completed 840 units against a target of 1,000."

NAME THE MEASURE

"Eighty-four percent is output target attainment."

STATE THE LOGGED LOSS

"The log records 40 minutes of downtime."

LIMIT ATTRIBUTION

"We have not established how much of the shortfall the stop explains."

QUANTIFY THE GAP

"The shortfall is 160 units, or 16% of target."

CORRECT THE LABEL

"The supplied figures do not support an OEE calculation."

KEEP CAUSE OPEN

"The reason for that downtime is still being checked."

ASK FOR THE TIME BASIS

"What planned production time was used in this calculation?"

Against

Introduces the comparison target, not necessarily customer demand.

Down by

State the amount and comparison period to avoid ambiguity.

Percent of

Identifies a ratio; it is not the same as a percentage-point change.

Phrases you can actually use

C Notice the language / 2 of 2

DISTINGUISH RATE FROM QUANTITY

"Throughput needs a quantity and a time period."

REQUEST THE BREAKDOWN

"Separate speed, availability, and quality information."

MAKE THE UPDATE USEFUL

"Report the known gap and the next evidence check separately."

ASSIGN THE FOLLOW-UP

"Mei will reconcile the log and report the supported loss breakdown."

CHECK THE CONSTRAINT

"We have not yet confirmed which step limited total output."

AVOID BLAME

"The current log does not establish an operator cause."

BOUND THE RECOVERY CLAIM

"A recovery proposal is not a guarantee that the missing output will be recovered."

CLOSE THE BOARD UPDATE

"Keep the metric definition beside the percentage."

Explains

Makes a causal or accounting claim that needs support.

Logged

Means recorded, not necessarily investigated or verified.

Recover

Specify output, schedule, or service; one recovery may not imply another.

Choose the precise message

D Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

1. Which board entry is accurate?

- A** OEE 84%, with all losses explained.
- B** Availability 84% and first-pass yield 100%.
- C** Forty missing units because downtime was 40 minutes.
- D** 840 units completed; 84% of target; 160-unit shortfall.

2. Why is OEE not established?

- A** Downtime alone supplies every OEE input.
- B** Planned time, ideal cycle time, and quality information are not supplied.
- C** OEE is always identical to target attainment.
- D** A target of 1,000 automatically defines ideal cycle time.

3. A manager converts the 40-minute stop into the full 160-unit shortfall. What is missing from that attribution?

- A** A supported running-rate and loss breakdown linking the stop to the output gap
- B** The subtraction of 840 completed units from the 1,000-unit target
- C** Confirmation that 40 is smaller than the target of 1,000
- D** A relabeling of target attainment as availability

4. If a second shift produces 900 against a target of 1,000, what is its attainment?

- A** 90%, six percentage points above 84%
- B** 90%, exactly six percent above 84%
- C** 10%, because 100 units are missing
- D** 900%, because 900 units were produced

Answers and explanations: [page 85](#).

Eight hundred forty is not the whole story

E Conversations / turns 1-10

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

WORD + PHRASE BANK

visual board / downtime / availability / Loss attribution / Quality loss / recovery plan / shortfall / performance loss / target attainment / bottleneck

- 01 **Daniel:** Yesterday's board shows eight hundred forty units against a thousand. It labels eighty-four percent as OEE, but I cannot see the inputs behind that label.
- 02 **Mei:** The figure is ¹ _____. We divided completed output by the stated target; we did not calculate the availability, performance, and quality factors.
- 03 **Daniel:** Please change the heading before the morning meeting. What is the gap in units, and what can we actually say about the stop?
- 04 **Mei:** The ² _____ is one hundred sixty units, or sixteen percent of target. That is a quantity gap, not a duration and not an explanation of the cause.
- 05 **Daniel:** The stop log records forty minutes. Does the team believe that interruption accounts for all the missing units, or is the analysis still open?
- 06 **Mei:** The ³ _____ is logged, but its cause and contribution remain under review. We should not convert forty minutes into one hundred sixty lost units without the relevant basis.
- 07 **Daniel:** We would need to know the expected production rate and whether other losses occurred. The board currently makes a single interruption look like a complete explanation.
- 08 **Mei:** Exactly. ⁴ _____ needs supporting data. I will reconcile the timing, output, and other relevant records rather than attach the entire gap to the most visible event.
- 09 **Daniel:** Which inputs do you need for the equipment measure? Give me the missing fields so I can ask the line team for them.
- 10 **Mei:** Start with the basis for ⁵ _____. We need the planned production time and the relevant running time, not merely the output target and one stop duration.

The conversation, continued

E Conversations / turns 11-20

Continue with the word bank on page 10.

- 11 **Daniel:** The speed side also needs a clear reference. Actual output can fall short even while the equipment is running, so time and quantity should not be merged.
- 12 **Mei:** Yes. The ⁶ _____ assessment needs the defined ideal rate and actual running performance. The supplied briefing does not provide that basis, so the board cannot claim a completed result.
- 13 **Daniel:** And the completed-unit count does not tell us how many met the quality definition. We should not silently assume that every produced unit was good.
- 14 **Mei:** Correct. ⁷ _____ needs the appropriate output breakdown. A production count by itself cannot establish that no defects or other quality losses occurred.
- 15 **Daniel:** Someone has suggested the packing station is the limiting step. That may be worth checking, but I have not seen evidence showing it constrained the whole line.
- 16 **Mei:** Keep ⁸ _____ as an investigation question. A busy station or visible queue does not automatically identify the system's actual limiting step.
- 17 **Daniel:** For today's meeting, I need a concise update: result, known interruption, open questions, and responsibility for the next check. We can discuss recovery after that evidence is clearer.
- 18 **Mei:** I will update the ⁹ _____ accordingly. It will show the output gap and logged stop separately, with me responsible for reconciling the records and reporting the supported breakdown.
- 19 **Daniel:** Good. We can acknowledge the missed target without guessing why it happened. Any proposal to recover output should make its assumptions and constraints clear.
- 20 **Mei:** Agreed. The ¹⁰ _____ will remain a proposal until assessed and authorized. First, the board must use the right measure and keep the unfinished investigation visible.

ANSWERS / EXPLANATIONS: PAGE 86

1. target attainment | 2. shortfall | 3. downtime | 4. Loss attribution | 5. availability | 6. performance loss | 7. Quality loss | 8. bottleneck | 9. visual board | 10. recovery plan

Compare the right percentages

F Say it

A line produces 450 units against a target of 500. It logs 20 minutes of downtime. No cause or OEE inputs are supplied.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

downtime / unconfirmed / fifty / ninety

Lead: "Target attainment is ¹ _____ percent."

Planner: "The output shortfall is ² _____ units."

Lead: "Twenty minutes describes recorded ³ _____."

Planner: "The cause remains ⁴ _____."

ANSWERS / EXPLANATIONS: PAGE 86

1. ninety | 2. fifty | 3. downtime | 4. unconfirmed

Rehearse with a fixed script

1. Read the production review. State 840 against 1,000, 84% target attainment, and a 160-unit shortfall.

2. Swap roles. Keep the forty-minute stop separate from its unconfirmed cause and output effect; do not label 84% as OEE.

3. Read the corrected 450-unit transfer. State ninety-percent attainment and a fifty-unit gap without inventing a stop cause.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

The queue is for a shared tool

A Read the situation

Describe observed waiting, challenge unsupported blame, and define a bounded improvement trial without claiming savings in advance.

Improvement facilitator Ana and supervisor Marcus observe a one-hour assembly session. Eight waiting episodes for a shared tool total 24 operator-minutes; some episodes overlap, so this is not 24 minutes of line downtime. The supervisor calls the operators slow. The team proposes a one-bench point-of-use tool trial, but no trial result exists. Ana and Marcus must distinguish waiting from working pace, explain the time basis, and keep safety and quality checks unchanged in the proposal.

PEOPLE / Ana: Improvement facilitator | Marcus: Assembly supervisor

INVITE THE PEOPLE WHO DO THE WORK

A respectful observation describes the delay and asks how the process creates it. Operators can explain practical constraints that a meeting-room proposal misses. Their involvement is not permission to bypass controls; it helps the team design a realistic, reviewable improvement trial.

Briefing checks

Circle one answer in each item. Use the briefing.

1. What was observed?

- A** Twenty-four minutes of confirmed line downtime
- B** A proven slow working pace for every operator
- C** A successful completed tool trial
- D** Eight tool-waiting episodes totaling 24 operator-minutes

2. Why is 24 not necessarily line downtime?

- A** The line was proved to run without any delay.
- B** Some waiting episodes overlap, and the measure is operator-minutes.
- C** Operator-minutes never include waiting.
- D** Eight episodes always equal eight hours.

3. What is the proposed change?

- A** A guaranteed saving already verified
- B** A one-bench point-of-use tool trial with safety and quality checks unchanged
- C** Immediate removal of all inspection steps
- D** A completed factory-wide rollout

Answers and explanations: page 87.

The vocabulary that does the work

B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

lean manufacturing

An approach to improving customer value by reducing waste in the process.

IN USE apply lean manufacturing

waste

Resource use that does not add the defined value or support a necessary requirement.

IN USE identify process waste

waiting

Time when work cannot proceed because a needed input is unavailable.

IN USE reduce waiting time

motion

Movement of people within a task or process.

IN USE analyze unnecessary motion

transportation

Movement of material between locations or steps.

IN USE reduce unnecessary transportation

overproduction

Producing earlier or in greater quantity than needed.

IN USE prevent overproduction

overprocessing

Performing more work than the relevant requirement needs.

IN USE identify overprocessing

excess inventory

Material held beyond the justified operating need.

IN USE reduce excess inventory

kaizen

Focused, ongoing improvement of work processes.

IN USE run a kaizen event

gemba

The actual place where the work occurs.

IN USE observe at the gemba

value stream

The connected activities and information involved in delivering a product.

IN USE map the value stream

current-state map

A representation of how the process currently works.

IN USE create a current-state map

Precision check

Twenty-four operator-minutes across eight episodes averages three operator-minutes per episode. Overlapping waits mean that total is not automatically elapsed line downtime. State the observation period and time basis when reporting the result.

The vocabulary that does the work

B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

future-state map

A proposed representation of an improved process.

IN USE develop a future-state map

point-of-use storage

Keeping needed items near where they are used.

IN USE trial point-of-use storage

5S

Workplace organization through sort, set in order, shine, standardize, and sustain.

IN USE sustain 5S practices

standard work

The agreed method and conditions for performing a process.

IN USE update standard work

pull system

Replenishment or production triggered by downstream need.

IN USE design a pull system

kanban

A signal authorizing defined replenishment or production in a pull system.

IN USE use a kanban signal

work-cell layout

The arrangement of people, equipment, and material in a production cell.

IN USE assess the work-cell layout

operator-minute

One minute of one operator's time, accumulated across people if specified.

IN USE report operator-minutes

baseline observation

A measurement of the current process before a proposed change.

IN USE record a baseline observation

pilot trial

A limited test of a proposed change before broader adoption.

IN USE define a pilot trial

countermeasure

An action intended to address an identified problem.

IN USE test a countermeasure

sustainment

Maintaining an improvement over time through defined follow-up.

IN USE plan sustainment

Precision check

Point-of-use storage is a proposed countermeasure, not proven savings. A bounded trial should define what is compared while retaining required safety and quality controls. A successful local trial would still need review before broader adoption.

Phrases you can actually use

C Notice the language / 1 of 2

DESCRIBE THE DELAY

"Operators waited for the shared tool during eight observed episodes."

SEPARATE PACE FROM ACCESS

"Waiting for a tool does not by itself show slow working pace."

INVITE PRACTICAL DETAIL

"Ask the operators where the tool is needed and what constrains access."

STATE THE PROPOSAL

"Trial point-of-use storage at one bench."

NAME THE TIME BASIS

"The total is 24 operator-minutes, not 24 minutes of line downtime."

CHALLENGE THE ATTRIBUTION

"The observation does not establish that operator speed caused the delay."

DEFINE THE CURRENT STATE

"Record the present locations, demand, and waiting pattern."

BOUND THE CLAIM

"No improvement result is available yet."

Waste

A process concept, not a label for an employee.

Slow

Can confuse working pace with delay caused by missing inputs.

Operator-minutes

Adds individual time; it may exceed elapsed clock time when waits overlap.

Phrases you can actually use

C Notice the language / 2 of 2

PRESERVE CONTROLS

"Required safety and quality checks remain unchanged."

CHECK FOR DISPLACED PROBLEMS

"Confirm that the change does not create a shortage at another bench."

AVOID PREBOOKED SAVINGS

"Do not report proposed time savings as already achieved."

RETAIN OPERATOR INPUT

"Include the people who use the tool in the practical review."

COMPARE CONSISTENTLY

"Use the same time basis and comparable task conditions."

SEPARATE A TRIAL FROM ROLLOUT

"A one-bench test is not a factory-wide implementation."

MAKE THE REVIEW SPECIFIC

"Review the waiting observations and any new difficulties after the trial."

PLAN THE FOLLOW-THROUGH

"Agree how an accepted improvement would be maintained."

Trial

Tests a proposal without claiming the result beforehand.

Unchanged

Specify the controls that remain in force during the trial.

Sustained

Means an improvement persists, not merely that one test looked favorable.

Choose the precise message

D Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

1. Which observation is defensible?

- A** The entire line stopped for 24 minutes.
- B** The proposed trial saved 24 minutes already.
- C** Eight tool-waiting episodes totaled 24 operator-minutes during the hour.
- D** Operators were lazy for exactly 40% of the shift.

2. What is the average time per recorded episode?

- A** Twenty-four hours
- B** Forty percent of every operator's shift
- C** Three operator-minutes
- D** Eight operator-minutes

3. Which proposal has suitable boundaries?

- A** Remove checks to guarantee a faster result.
- B** Roll out everywhere before reviewing the local effect.
- C** Declare all waits eliminated before the trial.
- D** Trial tool placement at one bench, keep required controls, and compare waiting under comparable conditions.

4. Waiting falls at the trial bench while the shared tool is less available elsewhere. Which conclusion still needs evidence?

- A** The existence of a proposal to change the tool location
- B** Total benefit across the connected process, including any displaced waiting
- C** The fact that the observation came from a particular bench
- D** The original eight waiting episodes in the supplied baseline

Answers and explanations: [page 87](#).

The queue is for a shared tool

E Conversations / turns 1-10

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

WORD + PHRASE BANK

countermeasure / motion / current-state map / waiting / point-of-use storage / baseline observation / pilot trial / value stream / sustainment / operator-minute

- 01 **Marcus:** We watched the bench for an hour and recorded eight delays. The team says the operators are slow, but each delay seemed to involve waiting for the shared tool.
- 02 **Ana:** Then name the observed ¹ _____. The tool was unavailable at those moments; that does not establish that the operators worked slowly when they had what they needed.
- 03 **Marcus:** The sheet adds up to twenty-four minutes. I was going to call that twenty-four minutes of line downtime, although two operators sometimes waited at the same time.
- 04 **Ana:** The measure is ² _____. The total accumulates individual waiting time, so overlapping episodes must not be presented as the same amount of elapsed whole-line downtime.
- 05 **Marcus:** I will put the one-hour observation period beside that figure. We did not watch a full shift, so I cannot call it a shift total.
- 06 **Ana:** Exactly. This is our ³ _____. Record the task conditions and tool demand so the later comparison is meaningful rather than just another isolated total.
- 07 **Marcus:** The operators suggested a tool location beside the bench. It sounds practical, but we need to understand whether other stations depend on that same shared tool.
- 08 **Ana:** Create the ⁴ _____ with their input. Show the tool's movements and users, not just the bench we observed.
- 09 **Marcus:** That would also help us separate people walking to retrieve the tool from material being moved between work areas. Both happen, but they are different kinds of movement.
- 10 **Ana:** Yes. Unnecessary ⁵ _____ concerns the people performing the task. We should describe the actual movement rather than use a broad waste label that hides what needs to change.

The conversation, continued

E Conversations / turns 11-20

Continue with the word bank on page 19.

- 11 **Marcus:** For the trial, the suggestion is to keep an appropriate tool at this one bench. We have not decided that every work area needs a new tool arrangement.
- 12 **Ana:** That is a proposal for ⁶ _____. Check the practical requirements and shared demand before assuming that moving the tool solves the problem for the whole process.
- 13 **Marcus:** We can start with one bench and compare the waiting pattern. Required inspection and safety checks should stay in place, so the trial does not gain time by removing controls.
- 14 **Ana:** Make that explicit in the ⁷ _____ description. Use comparable task conditions and the same measure, and record any new difficulties rather than reporting only the favorable observations.
- 15 **Marcus:** Hold on, the slide already says twenty-four minutes saved. Nobody has run the trial. Can we change that before it reaches the review meeting?
- 16 **Ana:** Correct it to proposed ⁸ _____. We can explain what the change is intended to address without booking the hoped-for benefit as an achieved result.
- 17 **Marcus:** After the trial, we should ask whether a reduction here created a tool shortage elsewhere. A faster bench would not necessarily mean a better overall flow.
- 18 **Ana:** Review the broader ⁹ _____. Local gains matter, but a change that simply transfers waiting to another step may not improve delivery to the customer.
- 19 **Marcus:** If the trial supports the change, the team still needs an approved method and follow-up. Otherwise the tool could gradually return to the old shared location.
- 20 **Ana:** That is ¹⁰ _____. Update the agreed method through the proper process, clarify ownership, and monitor whether the improvement persists without weakening the required controls.

ANSWERS / EXPLANATIONS: PAGE 88

1. waiting | 2. operator-minute | 3. baseline observation | 4. current-state map | 5. motion | 6. point-of-use storage | 7. pilot trial | 8. countermeasure | 9. value stream | 10. sustainment

Add people-time, not clock-time

F Say it

Four overlapping waiting episodes total 12 operator-minutes in a 30-minute observation. A new tool location is proposed, but no trial has run.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

savings / operator-minutes / trial / three

Facilitator: "The accumulated waiting measure is ¹ _____."

Supervisor: "Average waiting per episode is ² _____ minutes."

Facilitator: "The proposed tool location still needs a ³ _____."

Supervisor: "We cannot report achieved ⁴ _____ yet."

ANSWERS / EXPLANATIONS: PAGE 88

1. operator-minutes | 2. three | 3. trial | 4. savings

Rehearse with a fixed script

1. Read the tool-waiting discussion. Say eight episodes and twenty-four operator-minutes within the one-hour observation.
2. Swap roles. Contrast accumulated people-time with line downtime; keep the one-bench trial proposed and controls unchanged.
3. Read the corrected four-episode transfer. State three operator-minutes per episode, with no achieved savings yet.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

One thousand units, three different states

A Read the situation

Reconcile output categories and explain yield, pending recovery, and scrap without double-counting or assuming a rework outcome.

Quality analyst Priya and production manager Ellis review 1,000 inspected units from one batch. Exactly 900 passed on the first inspection, 60 are awaiting authorized rework, and 40 have an approved scrap disposition. These categories are mutually exclusive and account for all 1,000 units. No reworked units have yet been accepted. A summary labels all 100 non-first-pass units as scrap and forecasts 960 good units as though they already exist. Priya must correct both the loss and recovery statements.

PEOPLE / Priya: Quality analyst | Ellis: Production manager

MAKE THE CATEGORIES RECONCILE

A clear quantity discussion starts with the population and the mutually exclusive states. Separate completed results from expected recovery. This prevents a commercially attractive forecast from entering the good-unit count and avoids exaggerating waste by labeling all unfinished work as irreversible loss.

Briefing checks

Circle one answer in each item. Use the briefing.

1. What is the first-pass yield?

A 94%

B 10%

C 90%

D 96%

2. How many units have approved scrap disposition?

A 100

B 60

C 900

D 40

3. How many reworked units are already accepted?

A 960

B None

C 60

D 100

Answers and explanations: [page 89](#).

The vocabulary that does the work

B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

defect

A product characteristic that fails an applicable quality requirement.

IN USE classify a defect

defective unit

A unit containing one or more defects under the stated definition.

IN USE count defective units

nonconforming unit

A unit that does not meet a specified requirement.

IN USE identify nonconforming units

scrap

Product designated not to be used for its original intended purpose.

IN USE record approved scrap

rework

Action intended to bring nonconforming product into conformity.

IN USE perform authorized rework

repair

Action making product acceptable for intended use without necessarily meeting every original requirement.

IN USE distinguish repair from rework

first-pass yield

The share of units meeting requirements on their first pass.

IN USE calculate first-pass yield

final yield

The share meeting the stated acceptance criteria after the defined processing stages.

IN USE state the final-yield basis

rework queue

Units waiting for the approved rework process.

IN USE monitor the rework queue

scrap rate

The scrapped quantity divided by the stated reference quantity.

IN USE report the scrap rate

disposition

An authorized decision about how identified material will be handled.

IN USE approve a disposition

acceptance status

Whether the required acceptance decision has been completed.

IN USE verify acceptance status

Precision check

900 first-pass units plus 60 awaiting rework plus 40 scrap equals 1,000 inspected units. First-pass yield is 90%; approved scrap is 4% of this population. The remaining 6% is pending rework, not already recovered output.

The vocabulary that does the work

B Find the words / 2 of 2

03

Read the definitions and say the common word combinations aloud.

reinspection

Inspection after a corrective operation or under a defined repeat process.

IN USE complete reinspection

material reconciliation

Accounting for all material quantities across defined states.

IN USE perform material reconciliation

mutually exclusive categories

Groups defined so that an item belongs to only one of them.

IN USE use mutually exclusive categories

double-counting

Including the same item more than once in a total.

IN USE prevent double-counting

defect count

The number of individual defects, potentially more than one per unit.

IN USE distinguish defect count

unit count

The number of separate items in the measured population.

IN USE verify the unit count

cost of poor quality

Costs arising because products or processes fail to meet requirements.

IN USE assess cost of poor quality

scrap cost

The defined cost assigned to scrapped product or material.

IN USE calculate scrap cost

rework labor

Labor used to bring product into conformity through rework.

IN USE record rework labor

recovery assumption

A premise about the quantity expected to become acceptable.

IN USE qualify the recovery assumption

hold status

A controlled state preventing use pending the required decision.

IN USE maintain hold status

quality escape

Nonconforming product that passes beyond its intended detection point.

IN USE investigate a quality escape

Precision check

A unit may contain several defects, so defect counts and defective-unit counts are not interchangeable. Likewise, forecast recovery and accepted reworked quantity are different. If all 60 later pass, 960 accepted units would be a conditional future result, not the present count.

Phrases you can actually use

C Notice the language / 1 of 2

03

DEFINE THE POPULATION

"The report covers 1,000 inspected units from this batch."

SEPARATE THE PENDING QUANTITY

"Sixty units await authorized rework."

RECONCILE THE CATEGORIES

"These three exclusive states account for the full batch."

KEEP RECOVERY CONDITIONAL

"If all 60 pass after rework, the accepted total would be 960."

STATE THE FIRST-PASS RESULT

"Nine hundred met requirements on the first inspection."

STATE THE SCRAP DECISION

"Forty units have approved scrap disposition."

CORRECT THE SCRAP TOTAL

"The 100 non-first-pass units are not all scrap."

REPORT THE CURRENT STATE

"No reworked units have been accepted yet."

Awaiting

Indicates a future action, not completed recovery.

Accepted

Requires the defined decision, not merely a planned repair.

Would

Signals a conditional result when the necessary event has not occurred.

Phrases you can actually use

C Notice the language / 2 of 2

03

NAME THE YIELD BASIS

"First-pass yield is 90% of the inspected population."

DISTINGUISH DEFECTS AND UNITS

"Two defects on one item are not two defective units."

SEPARATE COSTS

"Scrap cost and rework labor are different cost categories."

CORRECT THE FORECAST LABEL

"Mark 960 as conditional recovery, not actual good output."

NAME THE SCRAP BASIS

"The approved scrap quantity is 4% of these 1,000 units."

AVOID DOUBLE-COUNTING

"Move accepted reworked units between states rather than add them twice."

KEEP AUTHORIZATION VISIBLE

"A rework plan does not remove the need for the required acceptance decision."

CLOSE THE REPORT

"Show current accepted, pending, and scrap quantities separately."

First-pass

Excludes later recovery from the initial success measure.

Unit

One item; it may contain more than one defect.

Scrapped

A disposition state, not a synonym for every quality problem.

Choose the precise message

D Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

1. Which report reconciles the batch correctly?
 - A 960 accepted now and 100 scrap
 - B 900 accepted and 100 confirmed scrap
 - C 1,000 accepted because rework is planned
 - D 900 first-pass accepted, 60 awaiting rework, and 40 scrap

2. If all 60 later pass the required acceptance process, what would the accepted total be?
 - A 960, making the original first-pass yield 96%
 - B 1,060 because the rework count is added twice
 - C 900 because reworked units can never be accepted
 - D 960, while first-pass yield remains 90%

3. One unit has three defects and is included in the sixty awaiting rework. How should the unit reconciliation treat it?
 - A Three accepted units once the three corrective operations are planned
 - B One pending unit, with three defects recorded in the separate defect count
 - C Three pending units because each defect needs its own correction
 - D One pending unit plus one scrap unit until rework is completed

4. Which financial statement is supported without further data?
 - A Total poor-quality cost is exactly 100 dollars.
 - B Planned rework has no labor cost.
 - C Every non-first-pass unit has the same financial consequence.
 - D Scrap quantity and pending rework are known; their complete costs are not supplied.

Answers and explanations: [page 89](#).

One thousand units, three different states

E Conversations / turns 1-10

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

WORD + PHRASE BANK

double-counting / recovery assumption / scrap / reinspection / first-pass yield / rework / mutually exclusive categories / material reconciliation / rework queue / defect count

- 01 **Ellis:** The headline says one hundred units scrapped. The detail says nine hundred passed first time, sixty await rework, and forty are scrap. Which number should I report?
- 02 **Priya:** Forty is the approved ¹ _____. quantity. The other sixty have not been discarded; they are awaiting a different process with its outcome still open.
- 03 **Ellis:** Let me check the batch total: nine hundred plus sixty plus forty is one thousand. Those categories cover every inspected unit once.
- 04 **Priya:** Yes, they are ² _____. Keep the pending units out of both current accepted output and the scrap total.
- 05 **Ellis:** The dashboard also shows ninety-six percent first-pass yield. Someone added the sixty units they expect to recover, even though none has passed after rework yet.
- 06 **Priya:** The ³ _____ is ninety percent. Nine hundred passed at first inspection. A later recovery does not change that original first-pass result.
- 07 **Ellis:** Then the current accepted count is nine hundred. I can show a possible nine hundred sixty separately, but not as output already available.
- 08 **Priya:** Correct. Sixty remain in the ⁴ _____. Completion of the planned work and the required acceptance are still ahead of them.
- 09 **Ellis:** Would this forecast work: if all sixty pass after the authorized process, final accepted output would be nine hundred sixty?
- 10 **Priya:** That is a clear ⁵ _____. Keep the if and would; otherwise the forecast reads like a completed recovery.

The conversation, continued

E Conversations / turns 11-20

03

Continue with the word bank on page 28.

- 11 **Ellis:** The technician called the work repair in an email. Should I use that word in the report, or retain the process named in the record?
- 12 **Priya:** Retain ⁶ _____ here: restoring conformity to the requirements. Repair can have a different acceptance basis, so the terms are not interchangeable.
- 13 **Ellis:** When the work finishes, should I move the units straight into accepted output? The work-order status will tell me that the operation is complete.
- 14 **Priya:** Wait for the required ⁷ _____ and acceptance decision. A finished operation does not by itself establish that the units meet requirements.
- 15 **Ellis:** Another table lists three defects on one unit. I need to keep that from becoming three defective units in the batch reconciliation.
- 16 **Priya:** Exactly. The ⁸ _____ can be higher than the defective-unit count. One physical unit can contribute several separate defects.
- 17 **Ellis:** And when a unit is accepted after rework, it leaves the pending category. Otherwise our total would grow without any additional material.
- 18 **Priya:** That would be ⁹ _____. Move the status when the acceptance is recorded; do not leave the same unit in both balances.
- 19 **Ellis:** I will show nine hundred currently accepted, sixty pending, and forty scrap. Finance will need the cost basis before turning those quantities into a monetary loss.
- 20 **Priya:** Complete that ¹⁰ _____ first. The states are known; equal cost per scrap and rework unit is not one of our supplied facts.

ANSWERS / EXPLANATIONS: PAGE 90

1. scrap | 2. mutually exclusive categories | 3. first-pass yield | 4. rework queue | 5. recovery assumption | 6. rework | 7. reinspection | 8. defect count | 9. double-counting | 10. material reconciliation

Keep future recovery conditional

F Say it

A batch of 200 has 170 first-pass accepted units, 20 awaiting rework, and 10 approved scrap. No rework outcome is available.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

eighty-five / conditional / ten / twenty

Analyst: "First-pass yield is ¹ _____ percent."

Manager: "The pending rework quantity is ² _____."

Analyst: "Approved scrap is ³ _____ units."

Manager: "A total of 190 accepted units remains ⁴ _____."

ANSWERS / EXPLANATIONS: PAGE 90

1. eighty-five | 2. twenty | 3. ten | 4. conditional

Rehearse with a fixed script

1. Read the batch review. Reconcile nine hundred first-pass accepted, sixty pending rework, and forty scrap to one thousand.
2. Swap roles. Keep ninety-percent first-pass yield unchanged by possible later recovery; 960 accepted remains conditional.
3. Read the corrected 200-unit transfer. State 85% first-pass yield, twenty pending, ten scrap, and 190 only if recovery succeeds.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

The supplier changed, but so did the process

04

A Read the situation

Challenge a premature root-cause statement, preserve competing hypotheses, and distinguish containment from verified corrective action.

Process engineer Jun and quality lead Rosa investigate a fictional surface defect. It first appears in the records after a supplier switch, but two process settings also changed during the same period. Comparable lot and setting records have not yet been reconciled. A draft names the supplier as the confirmed root cause and calls temporary sorting a completed corrective action. Jun and Rosa must organize the evidence, keep the cause open, and distinguish immediate screening from work that prevents recurrence.

PEOPLE / Jun: Process engineer | Rosa: Quality lead

A HYPOTHESIS IS NOT AN ACCUSATION

Use conditional language to keep plausible explanations available for testing. This is not a refusal to decide; it identifies what evidence a decision requires. A supplier discussion can be direct about the observed defect without presenting timing alone as proof of responsibility.

Briefing checks

Circle one answer in each item. Use the briefing.

1. What changed in the relevant period?

- A** No identified process input
- C** Only the supplier

- B** The supplier and two process settings
- D** Only the inspection report's color

2. What evidence is still unreconciled?

- A** Comparable lot and process-setting records
- C** A final supplier admission

- B** A completed proof that every setting was constant
- D** An approved effectiveness result

3. What is the status of temporary sorting?

- A** A demonstrated permanent cause elimination
- C** A completed effectiveness check for every future lot

- B** Evidence that the supplier definitely caused the defect
- D** An immediate screening measure, not proof of recurrence prevention

Answers and explanations: page 91.

The vocabulary that does the work

B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

root cause

An underlying cause whose removal addresses the problem's recurrence.

IN USE establish the root cause

symptom

An observable sign of a problem rather than its underlying explanation.

IN USE distinguish a symptom from a cause

hypothesis

A proposed explanation that can be examined against evidence.

IN USE test a hypothesis

correlation

An observed association that does not by itself establish causation.

IN USE distinguish correlation from causation

confounding factor

Another changing variable that can affect an apparent relationship.

IN USE identify confounding factors

change history

A record of relevant modifications over time.

IN USE reconstruct the change history

process parameter

A controllable or measured variable describing process operation.

IN USE review process parameters

lot traceability

The ability to link material and records to particular lots.

IN USE maintain lot traceability

comparative evidence

Information allowing relevant conditions or groups to be compared.

IN USE obtain comparative evidence

stratification

Separating data into meaningful groups for analysis.

IN USE stratify by supplier and setting

cause-and-effect diagram

A structured display of possible contributors to a problem.

IN USE build a cause-and-effect diagram

five whys

A questioning method used to explore successive levels of explanation.

IN USE use five whys carefully

Precision check

After the supplier switch describes timing, not a proven causal relationship. Two settings also changed, so those variables can confound the apparent supplier effect. Reconcile comparable records before treating one explanation as established.

The vocabulary that does the work

B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

8D

An eight-discipline structured approach to team problem solving.

IN USE prepare an 8D report

containment action

An immediate measure intended to limit the impact of a problem.

IN USE implement a containment action

sorting

Separating items using defined inspection or classification criteria.

IN USE document temporary sorting

escape point

The point at which a problem should have been detected but was not.

IN USE identify the escape point

occurrence cause

A cause explaining why the defect was produced.

IN USE verify the occurrence cause

detection failure

A reason an existing check did not identify the problem.

IN USE investigate a detection failure

corrective action

Action addressing a cause to prevent recurrence.

IN USE implement corrective action

effectiveness check

Evaluation of whether an action achieved its intended result.

IN USE define an effectiveness check

recurrence

The same or a related problem appearing again.

IN USE monitor recurrence

verification evidence

Recorded support that a specified condition or action has been achieved.

IN USE retain verification evidence

controlled comparison

A planned comparison that accounts for relevant differing conditions.

IN USE authorize a controlled comparison

closure criteria

Defined conditions for accepting an issue as resolved.

IN USE agree closure criteria

Precision check

Temporary sorting can limit escape of affected output without preventing the defect from being produced. A completed corrective action needs an identified cause, appropriate action, and the required effectiveness evidence. An analysis diagram organizes possibilities; it does not prove them.

Phrases you can actually use

C Notice the language / 1 of 2

STATE THE TIMING

"The defect first appears after the supplier switch in the available records."

QUALIFY THE EXPLANATION

"The supplier is one hypothesis, not a confirmed root cause."

SEPARATE ASSOCIATION

"The timing suggests a question; it does not establish causation."

DISTINGUISH PREVENTION

"Sorting does not by itself prevent the defect from being produced."

IDENTIFY COMPETING CHANGES

"Two process settings changed during the same period."

REQUEST COMPARABLE RECORDS

"Link the defect data to the actual supplier lots and settings."

DESCRIBE CONTAINMENT

"Temporary sorting screens output while the investigation continues."

KEEP THE EVIDENCE ORGANIZED

"Stratify the observations by the relevant conditions."

After

Indicates sequence, not automatically cause.

Because

Commits to an explanation that needs evidence.

Suspected

Keeps a plausible explanation open without declaring it confirmed.

Phrases you can actually use

C Notice the language / 2 of 2

AVOID AN UNSUPPORTED ACCUSATION

"Ask for supplier records without stating a cause we have not established."

BOUND FURTHER WORK

"Any comparison or trial must follow the authorized process."

DEFINE EFFECTIVENESS

"Specify what evidence would show the action prevents recurrence."

ASSIGN THE NEXT REVIEW

"Jun will reconcile the change and lot records for the next review."

CHECK DETECTION SEPARATELY

"Why did the existing check fail to identify the issue earlier?"

DO NOT PROMOTE A DIAGRAM

"A listed possible cause is not a verified finding."

KEEP CLOSURE OPEN

"The issue remains open until the agreed criteria are met."

CLOSE WITH STATUS

"Containment is active; cause and corrective-action effectiveness remain unconfirmed."

Contained

Means impact is being limited, not necessarily that the cause is removed.

Verified

Must identify which proposition or action was checked.

Closed

Requires the agreed criteria, not merely a completed meeting or form.

Choose the precise message

D Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

1. Which root-cause statement fits the evidence?

- A** The supplier is proven responsible because its change came first.
- B** The settings are proven responsible because there are two of them.
- C** No investigation is possible when several things change.
- D** The supplier change and two setting changes remain competing explanations pending comparison.

2. What does sorting establish by itself?

- A** The occurrence cause is eliminated.
- B** Every future lot will be defect-free.
- C** The supplier admitted responsibility.
- D** A screening activity is being used; recurrence prevention is not demonstrated.

3. Which request best tests the competing supplier and setting explanations?

- A** Group every later defect under the new supplier without checking setting history.
- B** Count all proposed causes and select the one mentioned most often.
- C** Treat completion of sorting as verification that the supplier caused the defect.
- D** Link defect observations to actual supplier lots and both settings under comparable conditions.

4. What distinguishes an occurrence cause from a detection failure?

- A** One explains production of the defect; the other explains why a check missed it.
- B** Both always mean the supplier changed.
- C** A missed check proves why the defect was created.
- D** Eliminating a detection failure automatically eliminates occurrence.

Answers and explanations: [page 91](#).

The supplier changed, but so did the process

E Conversations / turns 1-10

04

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

WORD + PHRASE BANK

detection failure / occurrence cause / closure criteria / containment action / lot traceability /
confounding factor / change history / effectiveness check / hypothesis / correlation

- 01 **Rosa:** The draft says the new supplier is the root cause because the defect first appeared after the switch. I want to check whether the evidence really isolates that change.
- 02 **Jun:** It does not yet. The supplier is a ¹ _____. Two process settings changed in the same period, and the comparable lot records have not been reconciled.
- 03 **Rosa:** I will keep the dates and remove the confirmed-cause label. Which competing explanations do you want the investigation team to check?
- 04 **Jun:** Exactly. The observed ² _____ does not establish which change produced the defect. We need to examine the competing explanations rather than select the easiest one to describe.
- 05 **Rosa:** The process settings could affect the result directly or interact with the supplied material. The current summary treats them as background details instead of relevant changes.
- 06 **Jun:** Each is a possible ³ _____. If conditions changed together, we cannot attribute the outcome to the supplier alone without further supporting comparison.
- 07 **Rosa:** What should the evidence package contain? The identifiers in our defect, receiving, and machine records do not yet line up.
- 08 **Jun:** Reconstruct the ⁴ _____ and link it to actual production. Dates in a meeting summary are not enough if the material or settings took effect at different times.
- 09 **Rosa:** Can you link each run to the actual material lot? The monthly supplier summary will not tell us whether the old stock was still being used.
- 10 **Jun:** That is where ⁵ _____ matters. Match the actual lots, settings, and observations before assuming that every unit made after the announcement used the new material.

The conversation, continued

E Conversations / turns 11-20

04

Continue with the word bank on page 37.

- 11 **Rosa:** The sorting team has been screening output while we investigate. The draft calls that corrective action complete, which sounds stronger than the work establishes.
- 12 **Jun:** Call it a ⁶ _____. Sorting may limit the movement of affected output, but it does not by itself stop the process from producing the defect.
- 13 **Rosa:** We should still evaluate the sorting method through the appropriate procedure. A temporary measure should not be treated as perfect simply because it is currently in place.
- 14 **Jun:** Agreed. And keep the ⁷ _____ separate from the screening question. One concerns why the defect is created; the other concerns whether the existing controls find it.
- 15 **Rosa:** That leaves a second investigation line: why the previous check did not flag the issue. Solving that could improve detection even before the production explanation is settled.
- 16 **Jun:** Yes, examine the ⁸ _____ on its own evidence. Do not assume that identifying a missed check also identifies the mechanism that created the defect.
- 17 **Rosa:** Once the evidence supports an action, we need a defined way to judge whether it worked. A completed supplier response form will not be enough.
- 18 **Jun:** Set an ⁹ _____ with the appropriate criteria and review period. Any trial or process change must follow the authorized procedure rather than become an improvised production experiment.
- 19 **Rosa:** I will revise the status: containment active, supplier and setting hypotheses open, and record reconciliation assigned to you. We will not announce a cause before the evidence supports it.
- 20 **Jun:** Good. Keep the ¹⁰ _____ visible so the team knows what remains. A clear open status is more useful than a closed label built on an untested explanation.

ANSWERS / EXPLANATIONS: PAGE 92

1. hypothesis | 2. correlation | 3. confounding factor | 4. change history | 5. lot traceability | 6. containment action | 7. occurrence cause | 8. detection failure | 9. effectiveness check | 10. closure criteria

Two changes, no isolated cause

F Say it

A defect appears after a material change and a temperature-setting change. No comparative analysis is complete. Temporary sorting is active.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

factor / evidence / hypothesis / action

Engineer: "The material explanation is still a ¹ _____."

Reviewer: "The setting change is a potential confounding ² _____."

Engineer: "Temporary sorting is a containment ³ _____."

Reviewer: "Recurrence prevention still needs effectiveness ⁴ _____."

ANSWERS / EXPLANATIONS: PAGE 92

1. hypothesis | 2. factor | 3. action | 4. evidence

Rehearse with a fixed script

1. Read the root-cause exchange. State the supplier switch and two setting changes without assigning a confirmed cause.
2. Swap roles. Distinguish sorting from recurrence prevention and the occurrence cause from a missed detection.
3. Read the corrected two-change transfer. Keep material and temperature hypotheses open while containment is active.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

A rush order meets the maintenance window

A Read the situation

Explain schedule conflicts, distinguish work types, and present conditional options without assuming authority to defer maintenance or restart equipment.

Maintenance planner Leila and production scheduler Ben face a conflict. A planned two-hour maintenance window runs from 13:00 to 15:00, while a new rush order would need the same line through 14:00 under the current schedule. No technical assessment or authorization to defer maintenance is supplied. An alternative line might be available, but its suitability is unconfirmed. Leila and Ben must prepare options for the responsible decision maker without describing a possible workaround as an approved plan.

PEOPLE / Leila: Maintenance planner | Ben: Production scheduler

ACKNOWLEDGE URGENCY WITHOUT BORROWING AUTHORITY

A useful maintenance discussion separates customer pressure from the technical assessment and the authorized decision. State the conflict precisely and present options with their conditions. Avoid a casual we can just move it when the people in the conversation have not established that deferral is acceptable.

Briefing checks

Circle one answer in each item. Use the briefing.

1. When do the two demands overlap?

- A** They do not overlap
- B** From 13:00 to 14:00
- C** From 14:00 to 15:00 only
- D** For the entire day

2. What is known about maintenance deferral?

- A** It is automatically safe because the order is urgent.
- B** Production has already authorized it.
- C** Maintenance has no effect on reliability or safety.
- D** No supporting technical assessment or authorization is supplied.

3. What is the alternative-line status?

- A** Fully qualified and reserved
- B** Proven to need no changeover
- C** Already producing the order
- D** Possible but unconfirmed in suitability

Answers and explanations: [page 93](#).

The vocabulary that does the work

B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

preventive maintenance

Planned work intended to reduce the likelihood of equipment failure.

IN USE schedule preventive maintenance

predictive maintenance

Maintenance informed by indicators of equipment condition or future failure.

IN USE use predictive maintenance

corrective maintenance

Work performed to restore function after a fault is identified.

IN USE plan corrective maintenance

condition monitoring

Tracking indicators of equipment health over time.

IN USE review condition-monitoring data

maintenance window

A reserved period for planned maintenance activity.

IN USE protect the maintenance window

work order

A controlled record authorizing and describing a defined maintenance task.

IN USE review the work order

backlog

Approved or identified work that remains unfinished.

IN USE prioritize the maintenance backlog

critical asset

Equipment whose failure has significant operational or other consequences.

IN USE identify critical assets

reliability

The ability to perform the required function over a stated period under stated conditions.

IN USE assess equipment reliability

maintainability

The ease and speed with which equipment can be maintained or restored.

IN USE improve maintainability

mean time between failures

Average operating time between failures for the defined repairable population.

IN USE interpret mean time between failures

mean time to repair

Average repair time under the stated measurement definition.

IN USE state the mean-time-to-repair basis

Precision check

The conflict is one hour, from 13:00 to 14:00, but moving a two-hour maintenance task may have consequences beyond that overlap. Its duration does not establish that it can be shortened, split, or deferred without the required assessment.

The vocabulary that does the work

B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

failure mode

The way in which an item fails to perform its function.

IN USE identify a failure mode

wear indicator

A measured or observed sign of component deterioration.

IN USE monitor wear indicators

spare-part readiness

Availability of the correct parts needed for the planned work.

IN USE confirm spare-part readiness

changeover

Transitioning equipment from one product or setup to another.

IN USE plan the changeover

setup time

The time required to prepare equipment for a specified job.

IN USE measure setup time

SMED

Single-minute exchange of die: a method for reducing changeover time.

IN USE apply SMED principles

internal setup

Setup activity that requires the machine to be stopped.

IN USE identify internal setup tasks

external setup

Setup activity that can be completed while the machine is running.

IN USE separate external setup tasks

deferral

Postponement of a planned task subject to the required assessment and decision.

IN USE assess a maintenance deferral

return to service

The authorized restoration of equipment to operational use.

IN USE confirm return to service

capacity contingency

A conditional alternative for meeting production needs.

IN USE assess a capacity contingency

schedule dependency

A condition or predecessor affecting the timing of planned work.

IN USE identify schedule dependencies

Precision check

SMED targets single-digit-minute changeovers, not a universal one-minute task. It distinguishes internal and external setup. It does not authorize skipping maintenance or safety controls. An alternative line needs suitability and readiness checks before a production commitment.

Phrases you can actually use

C Notice the language / 1 of 2

STATE THE OVERLAP

"The order needs the line until 14:00, while maintenance starts at 13:00."

NAME THE OPEN ASSESSMENT

"We do not yet have a basis for deferring the maintenance."

PRESENT A CONDITIONAL OPTION

"An alternative line may be possible if suitability and readiness are confirmed."

SEPARATE WORK TYPES

"This is maintenance, not merely a product changeover."

KEEP DURATION DISTINCT

"A one-hour conflict does not mean a two-hour task can be completed in one hour."

RESPECT DECISION OWNERSHIP

"The responsible decision maker needs the technical assessment before approving a change."

CHECK MORE THAN SPACE

"An idle line is not automatically qualified for this order."

PRESERVE REQUIRED SCOPE

"Do not remove required work just to fit the schedule."

Can

May describe physical possibility, capability, or permission; make the meaning clear.

Available

Does not necessarily mean suitable, qualified, or ready.

Deferred

Indicates a changed schedule, not evidence that the underlying need disappeared.

Phrases you can actually use

C Notice the language / 2 of 2

REQUEST READINESS INFORMATION

"Confirm the work scope, required parts, personnel, and dependencies."

QUALIFY THE FINISH TIME

"The plan needs its conditions stated before we promise completion."

OFFER COMMERCIAL ALTERNATIVES

"The decision maker can also assess a revised delivery commitment."

COMMUNICATE THE EFFECT

"Tell affected teams which line and time assumptions changed."

AVOID A RELIABILITY GUARANTEE

"Recent operation does not prove the task can be postponed without consequence."

KEEP RETURN DISTINCT

"Finishing the work is not the same as authorized return to service."

RECORD THE ACTUAL DECISION

"Update the schedule only after the authorized choice is documented."

CLOSE WITH NEXT STEPS

"Present the options and unresolved conditions together."

Completed

State whether work, checks, or return-to-service authorization is complete.

Changeover

A job transition, not a substitute word for every maintenance activity.

Contingency

A conditional alternative rather than a confirmed allocation.

Choose the precise message

D Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

1. Which schedule statement is accurate?

- A** There is a one-hour overlap, but the maintenance scope remains two hours unless properly reassessed.
- B** The maintenance must take only one hour because the overlap is one hour.
- C** The order and maintenance never conflict.
- D** The rush order automatically cancels the maintenance.

2. Which alternative-line statement is best?

- A** A possible alternative is already an approved schedule.
- B** Assess suitability, setup, readiness, and allocation before committing the order.
- C** Any idle line can run any product.
- D** Availability alone proves qualification.

3. Which reply preserves authority?

- A** A planner's suggestion is sufficient return-to-service approval.
- B** We can present deferral for assessment, but it is not authorized in the current information.
- C** We can skip it because production has a deadline.
- D** No assessment is needed if the machine is currently running.

4. The maintenance operation is marked finished. What remains a separate status to confirm?

- A** Whether the original window lasted two hours
- B** Whether the order and maintenance once overlapped
- C** Whether the planner had mentioned an alternative line
- D** Required checks and authorized return to service

Answers and explanations: [page 93](#).

A rush order meets the maintenance window

E Conversations / turns 1-10

05

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

WORD + PHRASE BANK

changeover / capacity contingency / Return to service / spare-part readiness / work order / schedule dependency / deferral / maintenance window / External setup / reliability

- 01 **Ben:** The rush order needs the line until fourteen hundred. Maintenance is booked from thirteen hundred to fifteen hundred, so the current schedule cannot satisfy both plans.
- 02 **Leila:** The ¹ _____ overlaps the order by one hour. That does not mean the two-hour task can be shortened to one hour.
- 03 **Ben:** Could we move the work later? I need an option for the customer discussion, but I do not have the technical basis to say that postponement is acceptable.
- 04 **Leila:** Treat ² _____ as an option for assessment, not an approved decision. We need the responsible review before changing the commitment or telling the team to proceed.
- 05 **Ben:** The customer will ask why we cannot use a machine that ran yesterday. What can I say without suggesting that yesterday's run settles today's maintenance decision?
- 06 **Leila:** It does not establish ³ _____ for the proposed period. The task's purpose, equipment condition, and consequences need assessment rather than a conclusion based only on yesterday's operation.
- 07 **Ben:** What information should we bring to the decision maker? I can show the order timing and the impact on delivery, but the work scope needs your input.
- 08 **Leila:** We should review the ⁴ _____, required resources, and relevant dependencies. The schedule discussion must not silently delete part of the task to make the overlap disappear.
- 09 **Ben:** Line two may be free this afternoon. I will check its product capability and setup requirements before offering it as the alternative.
- 10 **Leila:** Call it a ⁵ _____. An unoccupied line is not automatically suitable, ready, or allocated to this order, so we cannot promise that route yet.

The conversation, continued

E Conversations / turns 11-20

Continue with the word bank on page 46.

- 11 **Ben:** It would need a different product setup. That is separate from the maintenance on the original line, even though both activities take equipment time.
- 12 **Leila:** Correct. A ⁶ _____ prepares the alternative line for the job. Its scope and timing need review; it does not replace the original maintenance requirement.
- 13 **Ben:** The improvement team has discussed moving preparation outside stopped-machine time. We should not assume every preparation activity can be moved that way.
- 14 **Leila:** Exactly. ⁷ _____ refers to eligible tasks that can occur while equipment runs. Required safety, quality, and operating controls still govern how the work is organized.
- 15 **Ben:** We also need the correct maintenance parts and people available. A calendar reservation alone does not demonstrate that the work can finish as planned.
- 16 **Leila:** Confirm ⁸ _____ and the other resources. Missing items could change the plan, and the decision maker needs those facts alongside the customer timing.
- 17 **Ben:** Our options are therefore conditional: assess the alternative line, assess a permitted schedule change, or discuss a revised delivery commitment. None is confirmed yet.
- 18 **Leila:** Yes. Show each ⁹ _____ and the responsible owner. That makes the trade-off visible without allowing the rush order to become an unsupported technical authorization.
- 19 **Ben:** Once the authorized choice is recorded, I will update the affected teams. I will also avoid describing the equipment as ready merely because the work is reported finished.
- 20 **Leila:** Good. ¹⁰ _____ remains a distinct status. The final update should identify the completed work, required checks, and actual authorization before making a production commitment.

ANSWERS / EXPLANATIONS: PAGE 94

1. maintenance window | 2. deferral | 3. reliability | 4. work order | 5. capacity contingency | 6. changeover | 7. External setup | 8. spare-part readiness | 9. schedule dependency | 10. Return to service

An option still has conditions

F Say it

Maintenance is planned from 10:00 to 12:00. An order needs the same equipment until 11:00. A second line is possible but its suitability is unconfirmed.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

suitability / contingency / deferral / one

Planner: "The schedules overlap by ¹ _____ hour."

Technician: "Moving maintenance would be a proposed ² _____."

Planner: "The second line is a capacity ³ _____."

Technician: "Before using it, confirm its ⁴ _____."

ANSWERS / EXPLANATIONS: PAGE 94

1. one | 2. deferral | 3. contingency | 4. suitability

Rehearse with a fixed script

1. Read the schedule discussion. Identify the 13:00-to-14:00 conflict within the two-hour maintenance window.
2. Swap roles. Present the other line as conditional, and distinguish proposed deferral from authorization.
3. Read the corrected morning transfer. Preserve the one-hour overlap and the alternative line's unconfirmed suitability.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

Stop at the unclear instruction

A Read the situation

Raise an ambiguous safety instruction clearly, distinguish shutdown from energy control, and confirm the authorized clarification route.

During a classroom safety exercise, operator Hana reads a training card saying switch off, then clean. Trainer Oscar confirms that the card does not identify the equipment, energy sources, or applicable isolation procedure. No actual equipment work is underway. Hana is not being authorized to service a machine. The exercise must stop at the ambiguous instruction and refer to the current approved equipment-specific procedure and qualified personnel. Hana and Oscar must clarify the communication without inventing an isolation sequence.

PEOPLE / Hana: Operator in a classroom exercise | Oscar: Safety trainer

06

A SHORT STOP MESSAGE CAN BE RESPECTFUL

Safety clarification works best when the concern is specific and the pause is unambiguous. Say which instruction is unclear and what needs confirmation. Do not bury the concern in an apology or turn uncertainty into a request to guess the missing step.

Briefing checks

Circle one answer in each item. Use the briefing.

1. Where is this exchange taking place?

- | | |
|---|--|
| A In a classroom safety exercise with no equipment work underway | B During an authorized live servicing operation |
| C At a completed restart inspection | D In a confirmed emergency rescue |

2. Which clarification is needed before this classroom card could support the intended technical discussion?

- | | |
|---|--|
| A A generic sequence assumed suitable for every machine | B Identified equipment, energy sources, and the current applicable approved procedure |
| C A statement that switching off is always equivalent to isolation | D A learner signature substituting for the missing equipment reference |

3. What must not be invented?

- | | |
|--|---|
| A An isolation sequence for an unidentified machine | B A clear statement that the card is ambiguous |
| C A request for the approved procedure | D A named clarification route |

Answers and explanations: page 95.

The vocabulary that does the work

B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

EHS

Environment, health, and safety functions and practices.

IN USE raise an EHS concern

hazard

A source or condition with the potential to cause harm.

IN USE identify a hazard

hazardous energy

Energy that can injure someone if unexpectedly released or applied.

IN USE recognize hazardous energy

energy isolation

Separation from energy sources using the applicable control method.

IN USE follow the energy-isolation procedure

lockout

Securing an energy-isolating device in the required safe position under the applicable procedure.

IN USE follow the lockout procedure

tagout

Use of a warning tag within an applicable energy-control system.

IN USE understand tagout limitations

stored energy

Energy remaining in a system after its normal supply is interrupted.

IN USE address stored energy

residual energy

Energy that remains after an initial control or shutdown action.

IN USE assess residual energy

shutdown

Stopping normal equipment operation.

IN USE distinguish shutdown from isolation

unexpected startup

Unintended equipment operation that can expose people to hazards.

IN USE prevent unexpected startup

authorized employee

A person assigned and trained for the relevant hazardous-energy-control work.

IN USE identify the authorized employee

affected employee

A person whose operation or work area is affected by energy-control activity.

IN USE inform affected employees

Precision check

Stopping normal operation is not the same as controlling every hazardous energy source. This classroom card does not identify a machine or an approved method. The correct language response is to pause and seek the applicable procedure, not supply missing operational steps.

The vocabulary that does the work

B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

equipment-specific procedure

Approved instructions applicable to identified equipment and work.

IN USE consult the equipment-specific procedure

verification of isolation

Confirmation of the required isolated state using the approved method.

IN USE require verification of isolation

stop-work concern

A safety concern requiring work to pause under the applicable process.

IN USE raise a stop-work concern

near miss

An event that could have caused harm but did not in the observed outcome.

IN USE report a near miss

job hazard analysis

Review of job steps, associated hazards, and appropriate controls.

IN USE complete a job hazard analysis

permit to work

A formal authorization process for defined work where required.

IN USE verify permit-to-work requirements

personal protective equipment

Protective items worn to reduce exposure to specified hazards.

IN USE select required personal protective equipment

hierarchy of controls

An ordered approach to choosing hazard-control measures.

IN USE apply the hierarchy of controls

safety data sheet

A document describing a chemical's hazards and handling information.

IN USE consult the safety data sheet

exclusion zone

An area with controlled access because of defined hazards or work.

IN USE respect the exclusion zone

read-back

Repeating critical information to check shared understanding.

IN USE request a read-back

restart authorization

The required permission to resume equipment operation.

IN USE confirm restart authorization

Precision check

A tag and a physical lock do not perform identical functions. Training participation does not by itself authorize a person for a particular task. Apply the actual energy-control program and assigned roles; no sentence in this exercise permits servicing or restart.

Phrases you can actually use

C Notice the language / 1 of 2

STATE THE CONCERN

"Stop here: the instruction does not identify the equipment or isolation procedure."

DISTINGUISH THE ACTIONS

"Switching off is not the same as verifying energy isolation."

REQUEST THE CONTROLLED SOURCE

"Use the current equipment-specific procedure."

AVOID IMPROVISATION

"I will not infer the missing sequence from a generic instruction."

KEEP THE PAUSE CLEAR

"We should not continue from this ambiguous card."

IDENTIFY MISSING CONTEXT

"Which equipment and approved procedure does this instruction refer to?"

RESPECT ASSIGNED ROLES

"The relevant authorized personnel must handle the required assessment and work."

KEEP TRAINING SEPARATE

"This classroom exercise does not authorize actual servicing."

06

Stop

A clear pause instruction; do not dilute it with several uncertain alternatives.

Off

May describe a control state without establishing isolation.

Safe

A broad conclusion that needs a defined basis and scope.

Phrases you can actually use

C Notice the language / 2 of 2

ASK FOR CONFIRMATION

"Please confirm the document identity and current revision."

AVOID A FALSE SAFE CLAIM

"We have no evidence here that any equipment is isolated."

NAME THE ESCALATION

"Oscar will refer the ambiguous card for correction through the training process."

SEPARATE RESTART

"A clarification does not itself authorize equipment restart."

USE READ-BACK

"I will repeat the clarified scope and next step to check understanding."

DO NOT SUBSTITUTE PROTECTION

"Protective equipment does not replace the required energy-control process."

PRESERVE THE ISSUE

"Record the unclear wording so it is not reused unchanged."

CLOSE THE EXERCISE

"We will resume the classroom task only with the appropriate reviewed material."

06

Authorized

Refers to assigned role and required competence, not mere presence.

Verify

Requires the approved method; it is not a guess or visual assumption.

Read back

Confirms shared understanding but does not replace technical verification.

Choose the precise message

D Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

1. Which response best fits the ambiguous card?

- A** Ask an unassigned learner to certify the machine safe.
- B** Pause and obtain the current applicable equipment-specific procedure through the authorized route.
- C** Guess a generic isolation sequence from memory.
- D** Continue because the words switch off guarantee isolation.

2. What does switching off establish by itself?

- A** Restart is authorized.
- B** Every possible energy source has been verified controlled.
- C** Normal operation may be stopped; complete energy isolation is not established.
- D** All stored energy is necessarily removed.

3. What can read-back confirm?

- A** That no approved procedure is needed
- B** Shared understanding of the clarified message, not the physical isolated state
- C** That every machine is safe to clean
- D** That the learner is automatically authorized

4. Which closing statement is accurate?

- A** The equipment is now isolated because the discussion ended.
- B** Every participant can now perform servicing independently.
- C** The unclear card is acceptable if someone signs it.
- D** The ambiguous classroom card is referred for correction; no equipment work or restart is authorized.

Answers and explanations: [page 95](#).

Stop at the unclear instruction

E Conversations / turns 1-10

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

WORD + PHRASE BANK

Verification of isolation / EHS / restart authorization / Tagout / Stored energy / authorized employee / stop-work concern / Shutdown / read-back / equipment-specific procedure

06

- 01 **Hana:** I want to stop at this line. The card says switch off, then clean, but it does not identify the machine or the procedure we should be using.
- 02 **Oscar:** That is a clear ¹ _____. We are in a classroom exercise, and we should not continue as though the missing equipment information were already known.
- 03 **Hana:** The phrase switch off sounds simple, but it could refer only to stopping normal operation. It does not tell me what happens to other energy sources.
- 04 **Oscar:** Correct. ² _____ is not the same as complete energy control. This card gives us no basis for declaring any machine isolated or safe for the proposed work.
- 05 **Hana:** There could also be energy remaining after a normal stop. I should not assume that the absence of movement proves there is nothing left that could cause harm.
- 06 **Oscar:** Exactly. ³ _____ must be addressed through the applicable process. We will not improvise that process for unidentified equipment during a language exercise.
- 07 **Hana:** Which machine is this card supposed to cover? Can we get its approved instructions before the class goes any further?
- 08 **Oscar:** We need the current ⁴ _____ and the relevant qualified personnel. The correct document and assigned roles matter as much as the words on the card.
- 09 **Hana:** I understand the need to ask, but I do not want my question to sound as though I am volunteering to perform the isolation myself.
- 10 **Oscar:** You are not. The ⁵ _____ has an assigned role and the required training. Participating in this discussion does not give you that authorization.

The conversation, continued

E Conversations / turns 11-20

Continue with the word bank on page 55.

- 11 **Hana:** Some learners also treat a tag as though it physically prevents the same actions as a lock. We should make the distinction clear without demonstrating an unapproved method.
- 12 **Oscar:** Yes. ⁶ _____ has particular functions and limitations under the applicable system. A warning tag must not be described as providing identical physical restraint to a lock.
- 13 **Hana:** Once the correct information is available, I can repeat the equipment identity and the next communication step. That would help confirm that I heard the clarification correctly.
- 14 **Oscar:** That is ⁷ _____. It checks our shared understanding; it does not replace the required technical process or demonstrate that the equipment has reached a particular state.
- 15 **Hana:** The worksheet asks me to mark the machine isolated. I am leaving that blank: we are in a classroom and have not checked any equipment.
- 16 **Oscar:** Correct. ⁸ _____ is a separate technical requirement under the applicable procedure. Our conversation does not establish that it has occurred.
- 17 **Hana:** I will keep the concern specific: missing equipment identity and procedure. The training card needs correction, rather than a handwritten guess about the missing steps.
- 18 **Oscar:** I will refer it through the ⁹ _____ and training process. We should preserve the unclear wording so the responsible people can review and correct the source.
- 19 **Hana:** Then we pause this part of the exercise until the reviewed material is available. No one should interpret the clarification as permission to start cleaning a machine.
- 20 **Oscar:** Exactly. It also provides no ¹⁰ _____. The outcome today is a clear concern, a proper referral, and an explicit boundary between classroom communication and actual equipment work.

06

ANSWERS / EXPLANATIONS: PAGE 96

1. stop-work concern | 2. Shutdown | 3. Stored energy | 4. equipment-specific procedure | 5. authorized employee | 6. Tagout | 7. read-back | 8. Verification of isolation | 9. EHS | 10. restart authorization

Clarification is not authorization

F Say it

A classroom card omits the machine identity and approved procedure. No live equipment work has occurred. The trainer will obtain reviewed material.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

procedure / exercise / missing / restart

Learner: "The equipment identity is ¹ _____."

Trainer: "We must obtain the applicable approved ² _____."

Learner: "This discussion remains a classroom ³ _____."

Trainer: "It does not authorize a ⁴ _____."

ANSWERS / EXPLANATIONS: PAGE 96

1. missing | 2. procedure | 3. exercise | 4. restart

Rehearse with a fixed script

1. Read the classroom exchange. Pause at the ambiguous card and request the identified equipment and applicable approved procedure.
2. Swap roles. Distinguish shutdown, communication read-back, and physical isolation verification without inventing operating steps.
3. Read the corrected transfer. Keep the activity a classroom exercise with no servicing or restart authorization.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

One dimension, two drawing revisions

A Read the situation

Compare measured results against explicit revision limits and resolve the governing requirement before assigning responsibility or release status.

Supplier-quality engineer Farah and supplier contact Colin review a component measured at 12.18 mm. The incoming report omits the drawing revision. The inspector used revision C, specifying 12.00 plus or minus 0.10 mm; the supplier used revision B, specifying 12.20 plus or minus 0.10 mm. Which revision governs the order has not been confirmed. The lot is held pending review. Farah and Colin must state both comparisons accurately, trace the order documents, and avoid declaring either party responsible before resolving the requirement.

PEOPLE / Farah: Supplier-quality engineer | Colin: Supplier quality contact

07

RESOLVE THE REFERENCE BEFORE ARGUING ABOUT COMPLIANCE

Two people can make internally consistent statements while using different documents. Put the requirement identity and revision beside the result, then trace the agreed order. This avoids treating a reference mismatch as dishonesty and keeps the discussion focused on evidence and responsibility.

Briefing checks

Circle one answer in each item. Use the briefing.

1. What is the revision C range?

A 12.10 to 12.30 mm

B Exactly 12.18 mm only

C 11.00 to 13.00 mm

D 11.90 to 12.10 mm

2. How does 12.18 mm compare with revision B's stated range?

A It lies within 12.10 to 12.30 mm.

B It exceeds 12.30 mm.

C It is below 12.10 mm.

D It determines that revision B governs the order.

3. What remains unconfirmed?

A Which drawing revision governs this order

B The reported measured value

C That the inspector used C

D That the supplier used B

Answers and explanations: [page 97](#).

The vocabulary that does the work

B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

engineering drawing

A controlled technical representation of product requirements.

IN USE consult the engineering drawing

drawing revision

The identified version of a technical drawing.

IN USE confirm the drawing revision

nominal dimension

The stated target or reference size.

IN USE identify the nominal dimension

tolerance

The permitted variation from a specified value.

IN USE apply the stated tolerance

upper specification limit

The largest value permitted by the stated requirement.

IN USE check the upper specification limit

lower specification limit

The smallest value permitted by the stated requirement.

IN USE check the lower specification limit

measured value

The result reported by the measurement process.

IN USE record the measured value

measurement unit

The scale used to express a measurement, such as millimeters.

IN USE state the measurement unit

inspection characteristic

A specified feature selected for inspection.

IN USE identify the inspection characteristic

inspection report

A record of examined features, methods, and results.

IN USE correct the inspection report

purchase order

The commercial document defining the ordered goods and referenced requirements.

IN USE review the purchase order

order acknowledgment

The supplier's confirmation of the order terms it received or accepted.

IN USE compare the order acknowledgment

Precision check

Revision B permits 12.10 to 12.30 mm; revision C permits 11.90 to 12.10 mm. The measured 12.18 mm is within the supplied B limits and 0.08 mm above C's upper limit. Neither comparison decides which revision governs the order.

The vocabulary that does the work

B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

applicable revision

The document version governing the specified work or order.

IN USE establish the applicable revision

document discrepancy

A mismatch between relevant records or references.

IN USE resolve a document discrepancy

controlled copy

A document copy maintained under the applicable version-control process.

IN USE obtain the controlled copy

superseded revision

A version replaced by a later one in the relevant document system.

IN USE identify a superseded revision

engineering change notice

A controlled communication describing an approved technical change.

IN USE trace the engineering change notice

effectivity

The point or population from which a change applies.

IN USE confirm change effectivity

supplier flow-down

Communication of applicable requirements through the supply chain.

IN USE verify supplier flow-down

receiving record

Documentation of material received and its identifying details.

IN USE trace the receiving record

lot identity

The identifier linking material to a specific batch.

IN USE preserve lot identity

acceptance basis

The requirements and evidence used for an acceptance decision.

IN USE state the acceptance basis

deviation request

A request for authorized departure from a specified requirement.

IN USE review a deviation request

release decision

The authorized determination that material may proceed to use or distribution.

IN USE document the release decision

Precision check

Latest in a drawing system and applicable to this order are not necessarily identical. Review the order, acknowledgment, change communication, and effectivity. A numerical pass against one reference does not itself authorize release when the governing requirement remains unresolved.

Phrases you can actually use

C Notice the language / 1 of 2

STATE THE RESULT FULLY

"The reported dimension is 12.18 millimeters."

SHOW BOTH BASES

"The inspector used C; the supplier used B."

READ THE B LIMITS

"Revision B permits 12.10 to 12.30 millimeters."

AVOID PREMATURE BLAME

"We have not established which revision governs the order."

NAME THE MISSING REFERENCE

"The inspection report does not identify the drawing revision."

READ THE C LIMITS

"Revision C permits 11.90 to 12.10 millimeters."

REPORT THE COMPARISON

"The result is within B's stated limits and above C's upper limit."

TRACE THE AGREEMENT

"Compare the purchase order with the supplier acknowledgment."

Within

Names a comparison to defined limits; it does not identify which limits apply.

Latest

Describes version chronology, not necessarily contractual effectivity.

Applicable

Connects a requirement to the particular work, order, or lot.

Phrases you can actually use

C Notice the language / 2 of 2

CHECK THE CHANGE TIMING

"Confirm when revision C became applicable to this order or lot."

KEEP THE HOLD VISIBLE

"The lot remains held pending the required review."

REQUEST THE CONTROLLED SOURCE

"Send the referenced drawing and relevant change communication."

RECORD THE RESOLUTION

"Document the applicable revision and the resulting assessment."

PRESERVE THE ORIGINAL RECORD

"Add the corrected reference through the controlled process."

SEPARATE DIMENSION AND RELEASE

"Meeting one dimensional range is not the complete release decision."

CLARIFY A DEPARTURE

"A deviation request is not already an approved exception."

CLOSE THE SUPPLIER EXCHANGE

"Resolve the requirement basis before assigning responsibility or promising release."

Plus or minus

Sets symmetric limits around the stated nominal value.

Acknowledged

Records the supplier's response; compare it with the actual order terms.

Released

Requires the relevant authorization, not merely a favorable number.

Choose the precise message

D Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

1. What is the difference between 12.18 mm and revision C's upper limit?

- A** 0.10 mm below the limit
- B** 0.08 mm above the limit
- C** 0.18 mm above the limit
- D** 0.02 mm below the limit

07

2. Which update keeps the numerical result separate from the unresolved governing requirement?

- A** C is the later letter, so it automatically governed every earlier order.
- B** 12.18 meets B's 12.10-12.30 range and exceeds C's upper limit by 0.08; applicability and release remain open.
- C** 12.18 meets B, so B must be the applicable requirement and the lot can be released.
- D** 12.18 exceeds C's nominal by 0.18, so it exceeds C's upper limit by 0.18.

3. Which evidence best addresses the mismatch?

- A** A verbal claim that both revisions are equivalent
- B** Order references, acknowledgment, drawing versions, and change effectivity records
- C** Only the most recent filename in someone's downloads
- D** The supplier's production deadline alone

4. How should the report be corrected?

- A** Delete the original record and release immediately.
- B** Remove units and tolerances to avoid disagreement.
- C** Add the missing revision and resolution through the controlled record process while retaining traceability.
- D** Replace the result silently so it appears to meet C.

Answers and explanations: [page 97](#).

One dimension, two drawing revisions

E Conversations / turns 1-10

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

WORD + PHRASE BANK

supplier flow-down / deviation request / measured value / release decision / order acknowledgment / effectivity / lower specification limit / acceptance basis / upper specification limit / drawing revision

07

- 01 **Colin:** We measured 12.18 millimeters and reported it within tolerance. Your incoming report says it failed. Can we compare the requirement before we debate the result?
- 02 **Farah:** Yes. The report is missing the ¹ _____. Your team used B, our inspector used C, and we have not confirmed which governs this order.
- 03 **Colin:** B says nominal 12.20, plus or minus 0.10 millimeters. We used that drawing in production, but I agree its applicability still needs checking.
- 04 **Farah:** B's ² _____ is 12.10 millimeters, and its upper limit is 12.30. The measured 12.18 is inside that interval.
- 05 **Colin:** C says 12.00, plus or minus 0.10. The upper end looks like the lower end of B, which may be why the report became confusing.
- 06 **Farah:** C's ³ _____ is 12.10. The measurement is 0.08 above it, with C's full range running from 11.90 to 12.10.
- 07 **Colin:** Then neither team needs a different measurement to explain the disagreement. The same 12.18 gets a different comparison because the references differ.
- 08 **Farah:** Keep the ⁴ _____ unchanged. We need to resolve the reference, not adjust the recorded number until it fits a preferred drawing.
- 09 **Colin:** Our production team says B came with the order. I will send the actual package rather than ask you to accept that recollection.
- 10 **Farah:** Include the ⁵ _____ and attached drawing. I will compare them with our purchase order and the controlled requirement references.

The conversation, continued

E Conversations / turns 11-20

Continue with the word bank on page 64.

- 11 **Colin:** I found a later change notice too. Its issue date does not tell me whether it applied to material already ordered.
- 12 **Farah:** Check its ⁶ _____: the order, lot, date, or other stated application point. Newest in the system does not necessarily mean governing this lot.
- 13 **Colin:** If C was meant to apply, we also need to know whether the requirement reached our team. I will trace the communication history.
- 14 **Farah:** That is the ⁷ _____ question. Keep it evidence-based; a mismatch does not yet tell us which party caused it.
- 15 **Colin:** Production wants a release estimate. Can I say the lot is usable because 12.18 meets B, while we finish the paperwork?
- 16 **Farah:** No. The ⁸ _____ is unresolved and the lot is held. A favorable comparison with B does not supply the missing authorization.
- 17 **Colin:** We could request a formal exception if one is needed. I will keep that separate from asking which revision actually governed the order.
- 18 **Farah:** Correct. A ⁹ _____ is a request, not an approved departure. Its submission would not itself remove the hold.
- 19 **Colin:** My update will show both ranges, the unchanged measurement, and the documents we are tracing. I will not promise use of the lot.
- 20 **Farah:** Once the requirement is resolved, record the authorized ¹⁰ _____ and correct the report transparently. Preserve the measurement, references, and history of the discrepancy.

07

ANSWERS / EXPLANATIONS: PAGE 98

1. drawing revision | 2. lower specification limit | 3. upper specification limit | 4. measured value | 5. order acknowledgment | 6. effectivity | 7. supplier flow-down | 8. acceptance basis | 9. deviation request | 10. release decision

Pass against which revision?

F Say it

A measurement is 5.08 mm. Revision A allows 4.90-5.10 mm; revision B allows 4.95-5.05 mm. The governing revision is unconfirmed and the lot remains held.

Complete the four gaps. Use each bank entry once.

07

WORD + PHRASE BANK

limit / unconfirmed / A / released

Engineer: "The result is within revision ¹ _____ limits."

Supplier: "It exceeds revision B's upper ² _____."

Engineer: "The applicable revision remains ³ _____."

Supplier: "The lot has not been ⁴ _____."

ANSWERS / EXPLANATIONS: PAGE 98

1. A | 2. limit | 3. unconfirmed | 4. released

Rehearse with a fixed script

1. Read the supplier exchange. Compare 12.18 with B's 12.10-12.30 and C's 11.90-12.10 millimeter ranges.
2. Swap roles. State 0.08 above C's upper limit; preserve the measurement and the unresolved order effectivity.
3. Read the corrected 5.08-millimeter transfer. It meets A and exceeds B; applicability is unconfirmed and the lot remains held.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

Six checks complete, two still open

A Read the situation

Transfer unfinished work with specific identifiers, evidence, ownership, dependencies, and a confirmed next update.

Outgoing supervisor Diego hands lot L27 to incoming supervisor Elena. Six of eight required release checks are complete, with records attached. Check 7, label-count reconciliation, remains open; check 8, final quality review, depends on it. Site procedure requires all eight before lot release. Elena is the named incoming supervisor but has not yet confirmed receipt. A draft handoff says 75% ready and should be fine. Diego and Elena must replace that reassurance with a precise status and a closed-loop handoff.

08

PEOPLE / Diego: Outgoing supervisor | Elena: Incoming supervisor

OWNERSHIP IS NOT COMPLETION

An incoming supervisor can accept responsibility for follow-up without declaring the work finished. A good read-back identifies the open tasks and restrictions, not just the phrase got it. This matters especially when a friendly handoff risks turning an optimistic expectation into an apparent release decision.

Briefing checks

Circle one answer in each item. Use the briefing.

1. Which checks remain open?

A Check 7 label-count reconciliation and check 8 final quality review

C None because six are complete

B All eight checks

D Only an unspecified optional check

2. What is required before lot release?

A Merely naming the incoming supervisor

C A 75% completion percentage alone

B Completion of all eight required checks under the site procedure

D A friendly verbal reassurance

3. What has Elena not yet confirmed at the start?

A That Diego is on the outgoing shift

C That eight checks exist

B That the lot is called L27

D Receipt of the handoff

Answers and explanations: page 99.

The vocabulary that does the work

B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

shift handoff

Transfer of relevant work status and responsibility between shifts.

IN USE conduct a shift handoff

outgoing shift

The team ending its scheduled work period.

IN USE brief the outgoing shift

incoming shift

The team beginning responsibility for the next period.

IN USE inform the incoming shift

open item

A task or issue that has not met its completion criteria.

IN USE identify open items

completed check

A required check with its completion evidence recorded.

IN USE verify completed checks

release checklist

The defined checks required for a material or product release decision.

IN USE review the release checklist

label reconciliation

Accounting for issued, used, unused, and rejected labels as required.

IN USE complete label reconciliation

final quality review

The designated review of the required evidence before a release decision.

IN USE complete the final quality review

dependency

A task or condition that must be resolved before another can proceed.

IN USE identify a task dependency

handoff record

The documented status and responsibilities transferred between teams.

IN USE update the handoff record

attachment

A supporting file or record linked to the handoff.

IN USE verify the attachment

record reference

An identifier locating the supporting evidence.

IN USE provide the record reference

Precision check

Six of eight checks is 75% by count, but it is not 75% release authorization or a measure of remaining risk. The supplied procedure requires all eight. Name the two open checks and keep the lot-release restriction explicit.

The vocabulary that does the work

B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

named owner

A specifically identified person responsible for an action.

IN USE assign a named owner

acknowledgment

Confirmation that a message was received, distinct from accepting responsibility.

IN USE obtain acknowledgment

closed-loop communication

An exchange in which receipt and understanding are confirmed.

IN USE use closed-loop communication

read-back

Repetition of key information to verify shared understanding.

IN USE request a read-back

next checkpoint

The agreed time or event for the next status review.

IN USE set the next checkpoint

due time

The time by which a task is expected or required to be complete.

IN USE distinguish due time from update time

escalation trigger

A condition requiring referral to the next responsible level.

IN USE define an escalation trigger

handover acceptance

Confirmation of responsibility for the transferred follow-up.

IN USE record handover acceptance

status qualifier

A word or phrase defining the limits of a reported status.

IN USE preserve status qualifiers

pending decision

A required determination that has not yet been made.

IN USE flag a pending decision

continuity

Preservation of information and control across changes of personnel.

IN USE maintain continuity

audit trail

A traceable history of records, actions, and decisions.

IN USE preserve the audit trail

Precision check

An update time is not a guaranteed completion time. Receipt, ownership, task completion, and release are distinct statuses. The final quality review depends on label reconciliation, so assigning both tasks does not remove their sequence.

Phrases you can actually use

C Notice the language / 1 of 2

IDENTIFY THE LOT

"This handoff concerns lot L27."

NAME EACH OPEN ITEM

"Check 7 is label reconciliation; check 8 is final quality review."

KEEP RELEASE EXPLICIT

"The lot is not released while the required checks remain open."

NAME THE RECIPIENT

"Elena is the incoming supervisor responsible for follow-up."

STATE COMPLETED WORK

"Six required checks are complete, with the records attached."

DESCRIBE THE DEPENDENCY

"The final review depends on completion of the reconciliation."

REPLACE VAGUE REASSURANCE

"Please remove should be fine from the status."

CONFIRM RECEIPT

"Please confirm you can access the attached records."

08

Ready

Specify ready for what: handoff, review, or release.

Done

Needs an identified task and supporting completion evidence.

Should be fine

Reassurance that does not describe actual status or restrictions.

Phrases you can actually use

C Notice the language / 2 of 2

REQUEST A PRECISE READ-BACK

"Repeat the two open checks and the release restriction."

SET AN UPDATE

"Agree the next checkpoint without promising an unverified finish time."

PRESERVE REFERENCES

"Keep the record identifiers beside the completed checks."

RECORD CHANGES

"Update each status when supporting evidence becomes available."

SEPARATE OWNERSHIP

"Accepting the handoff does not mean the checks are complete."

DEFINE ESCALATION

"If required evidence is unavailable, follow the site's escalation route."

AVOID A MISLEADING PERCENTAGE

"Seventy-five percent counts checks; it does not authorize release."

CLOSE THE LOOP

"Confirm the handoff, owner, open items, and next communication."

08

Accepted

May mean ownership accepted, not product accepted; name the object.

By

Can imply a deadline; do not use it for an update unless that meaning is clear.

Pending

Preserves an unresolved task or decision rather than predicting its outcome.

Choose the precise message

D Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

1. Which handoff line is most useful?

- A** L27: checks 1-6 complete; 7 reconciliation and 8 final review open; lot not released.
- B** L27 is 75% ready and should be fine.
- C** All checks complete because the incoming supervisor is named.
- D** L27 released subject to two required checks that have not occurred.

08

2. Elena confirms receipt and explicitly accepts follow-up responsibility. What may the record state?

- A** Handoff received and ownership accepted; checks 7 and 8 remain open and the lot is not released.
- B** Receipt confirmed, so both remaining checks are technically complete.
- C** Ownership accepted, so final quality review can precede reconciliation.
- D** Handoff accepted, so release is authorized subject to later evidence.

3. The teams agree an update at 09:00. Which interpretation is safest and most precise?

- A** A status report is due at 09:00; completion is not guaranteed unless separately established.
- B** Both checks must already be complete because an update time exists.
- C** The lot is released at 09:00 regardless of results.
- D** No communication is needed unless everything finishes.

4. Which task order matches the brief?

- A** Complete label reconciliation before the dependent final quality review.
- B** Complete final review by assuming reconciliation will pass later.
- C** Mark both complete when the attachments are opened.
- D** Remove check 7 because six others are finished.

Answers and explanations: page 99.

Six checks complete, two still open

E Conversations / turns 1-10

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

WORD + PHRASE BANK

open item / handover acceptance / read-back / escalation trigger / attachment / audit trail / release checklist / dependency / status qualifier / next checkpoint

08

- 01 **Diego:** This handoff covers lot L27. Six of the eight required checks are complete, but the draft says seventy-five percent ready and should be fine.
- 02 **Elena:** Please replace that with the actual ¹ _____ list. A percentage by check count does not tell me which work remains or whether the lot can be released.
- 03 **Diego:** Check seven is label-count reconciliation. Check eight is the final quality review, and that review depends on the completed reconciliation.
- 04 **Elena:** I will preserve that ² _____. Assigning both tasks does not make them independent or allow the final review to assume a result that is still missing.
- 05 **Diego:** The completed-check records are attached, with identifiers next to checks one through six. I have named you as the incoming supervisor, but I still need confirmation.
- 06 **Elena:** First, let me verify each ³ _____ is accessible. A file listed in a message is not useful evidence if the incoming team cannot open it or identify what it supports.
- 07 **Diego:** Thank you. The site procedure requires all eight checks before release, so the two unfinished checks mean L27 has not been released.
- 08 **Elena:** That restriction belongs beside the ⁴ _____. Six completed items do not reduce the requirement to finish the other two, regardless of how reassuring the summary sounds.
- 09 **Diego:** I will delete should be fine. What exact status would you want the incoming team to see first?
- 10 **Elena:** Use a clear ⁵ _____ instead: not released, two checks open. That tells the team what the status means without suggesting a result we do not have.

The conversation, continued

E Conversations / turns 11-20

Continue with the word bank on page 73.

- 11 **Diego:** Can you confirm the records and the two remaining checks now? I want the handoff to be more than a message sent just before I leave.
- 12 **Elena:** Yes. Here is my ⁶ _____: L27, checks one through six complete; seven reconciliation and eight final review open; final review depends on seven; no release yet.
- 13 **Diego:** That matches. You are taking responsibility for coordinating the follow-up, but the record should not mark either check complete just because you accepted the handoff.
- 14 **Elena:** Correct. Record ⁷ _____ separately from check completion. I accept the follow-up responsibility; the actual check results and authorized decision still need their own evidence.
- 15 **Diego:** Can we agree on an update at nine tomorrow, even if the reconciliation is still open? I would rather hear the actual position than wait for good news.
- 16 **Elena:** Nine will be the ⁸ _____, not a guaranteed finish time. I will report the actual status and any unresolved issue rather than make completion a condition for communicating.
- 17 **Diego:** If a required record is unavailable, we need the team to use the site's escalation process instead of treating silence as permission to move the lot.
- 18 **Elena:** I will state that ⁹ _____ in the handoff. The appropriate route remains the site process; I will not invent an alternative approval because a record is delayed.
- 19 **Diego:** The corrected record now names the lot, evidence, two open checks, dependency, release restriction, owner, and update time. That is clearer than a readiness percentage.
- 20 **Elena:** It also preserves the ¹⁰ _____. Later updates can show when each check was completed and who made the decision, without changing what was actually open at this handoff.

08

ANSWERS / EXPLANATIONS: PAGE 100

1. open item | 2. dependency | 3. attachment | 4. release checklist | 5. status qualifier | 6. read-back | 7. handover acceptance | 8. next checkpoint | 9. escalation trigger | 10. audit trail

Acknowledged is not completed

F Say it

Lot M8 has three of four required checks complete. The last review is open. Noor accepts follow-up responsibility and agrees to provide an update at 16:00, with no completion guarantee.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

time / responsibility / release / open

08

Outgoing lead: "The final review remains ¹ _____."

Noor: "I accept follow-up ² _____."

Outgoing lead: "Sixteen hundred is the agreed update ³ _____."

Noor: "The handoff does not itself authorize ⁴ _____."

ANSWERS / EXPLANATIONS: PAGE 100

1. open | 2. responsibility | 3. time | 4. release

Rehearse with a fixed script

1. Read the L27 handoff. Name checks 7 and 8, their dependency, and the not-released status.
2. Swap roles. Confirm receipt and accept responsibility explicitly; make nine tomorrow an update time, not a completion promise.
3. Read the corrected M8 transfer. Keep the final review open and 16:00 as Noor's status update.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

The gauge passes today, but what about yesterday?

E More workplace conversations / 1-10

Report a measurement concern and define the scope of a product-impact review.

A micrometer fails its incoming calibration check and passes after adjustment. It was used to accept three production lots yesterday. The quality engineer and supervisor must organize a retrospective review. The fictional site has removed the instrument from production use pending the required decisions.

PEOPLE / Marta: Quality engineer | Cal: Production supervisor

Use each bank entry once, without changing its form. Answers follow turn 20.

09

WORD + PHRASE BANK

as-left results / reinspection / fitness for purpose / asset identifier / measurement uncertainty / as-found results

- 01 **Cal:** The lab adjusted the micrometer and it passes now. Can I close the issue and put yesterday's three lots back on schedule?
- 02 **Marta:** Not yet. I need the ¹ _____, not just the passing report after adjustment. What did it read before anyone changed it?
- 03 **Cal:** I have the final certificate here. The earlier readings are on a separate sheet; I will request that too.
- 04 **Marta:** Thank you. The ² _____ tell us its condition afterward. They do not resolve the earlier acceptance decisions.
- 05 **Cal:** We know it was used yesterday, but two operators shared it. Their inspection sheets should give us the part numbers.
- 06 **Marta:** Match those sheets to the instrument's ³ _____. We must not include every micrometer just because they look alike.
- 07 **Cal:** Should I tell the warehouse that all three lots are defective? They are asking what to put in their update.
- 08 **Marta:** Say acceptance is under review. A suspect measurement does not automatically prove that every part is defective.
- 09 **Cal:** What else will you need once we have the readings and the lot list?
- 10 **Marta:** The measured features, limits, method, and relevant ⁴ _____. Then we can assess which decisions may be affected.

The conversation, continued

E More workplace conversations / 11-20

Continue with the word bank on page 76.

- 11 **Cal:** Some results were near a limit. I will flag those, but I will not quietly replace them with fresh measurements.
- 12 **Marta:** Correct. Preserve the original records. Any ⁵ _____ needs an agreed plan, suitable equipment, and separate results.
- 13 **Cal:** One of the lots has already left the site. I will identify its destination rather than leave it out of the review.
- 14 **Marta:** Please do. Include shipped material in the impact assessment and route any required customer action to the responsible team.
- 15 **Cal:** Can we borrow another calibrated micrometer for the next job while you review these lots?
- 16 **Marta:** Check its range, resolution, condition, and suitability for the feature. A calibration label alone does not establish ⁶ _____.
- 17 **Cal:** Understood. I will send the three lot records and identify which instrument handled each inspection before the planning meeting.
- 18 **Marta:** I will bring the initial assessment and list anything still missing. Keep the affected material status visible until the required disposition is recorded.
- 19 **Cal:** The update will separate today's instrument condition from yesterday's product decisions. That should prevent the passing certificate from closing both questions.
- 20 **Marta:** Exactly. We need a defensible conclusion about the earlier measurements, not just a tool that reads correctly now.

09

ANSWERS / EXPLANATIONS: PAGE 101

1. as-found results | 2. as-left results | 3. asset identifier | 4. measurement uncertainty | 5. reinspection | 6. fitness for purpose

A replacement gauge is not automatically suitable

F Say it in a new situation

A replacement instrument has a current calibration label, but its range and measurement capability have not been checked against the required feature.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

limits / measurement / inspection / instrument

09

Supervisor: The label does not establish fitness for this ¹ _____.

Engineer: Compare the instrument capability with the feature and its ² _____.

Supervisor: Keep earlier results separate from any new ³ _____.

Engineer: Record the basis for selecting the replacement ⁴ _____.

ANSWERS / EXPLANATIONS: PAGE 101

1. measurement | 2. limits | 3. inspection | 4. instrument

Read, switch, repeat

1. Check the four answers and read the complete exchange aloud.
2. Switch roles. Read it again with the word bank covered.
3. Return to the extended conversation. Read turns 1-10, then switch roles for turns 11-20. Use the corrected expressions.

SOLO PRACTICE

Read both roles. Pause briefly at each change of speaker. Repeat any line that needed an answer check.

The right total, the wrong replenishment mix

E More workplace conversations / 1-10

Resolve a line-side material shortage without confusing quantities, variants, and replenishment signals.

An assembly cell needs 40 left-hand and 40 right-hand brackets. Its delivery contains 60 left-hand and 20 right-hand brackets. The total of 80 is correct, but only 20 complete pairs are available. A material handler and cell leader reconcile the delivery and the replenishment signal.

PEOPLE / Inez: Cell leader | Dev: Material handler

Use each bank entry once, without changing its form. Answers follow turn 20.

10

WORD + PHRASE BANK

shortage / pick list / kanban / part-number mix / Held stock / inventory balance

- 01 **Inez:** Dev, the trolley has eighty brackets, but we cannot build forty assemblies from it. Can you check the two variant labels with me?
- 02 **Dev:** I see the problem. The ¹ _____ is sixty left-hand and twenty right-hand, rather than forty of each.
- 03 **Inez:** Each assembly uses one of each. That gives us twenty complete pairs, with forty left-hand brackets left over.
- 04 **Dev:** So the immediate ² _____ is twenty right-hand brackets. I will locate those before asking stores for another eighty parts.
- 05 **Inez:** Thank you. The last delivery note only showed the family total. We need the two part numbers on the correction.
- 06 **Dev:** I will check the ³ _____ against what was actually loaded. We should not assume the error started in stores.
- 07 **Inez:** The line-side bins look similar, and one replenishment card was tucked behind the other. That may be relevant, but let us preserve the actual record.
- 08 **Dev:** Agreed. I will match each ⁴ _____ to its part number and standard quantity before changing anything.
- 09 **Inez:** Please do not increase both quantities as a quick fix. We have limited space, and extra left-hand stock will not solve the missing pairs.
- 10 **Dev:** No. I will return the excess through the normal process and correct the ⁵ _____ after the movement is recorded.

The conversation, continued

E More workplace conversations / 11-20

Continue with the word bank on page 79.

- 11 **Inez:** Can you give me a reliable arrival time for the twenty missing pieces? I need to coordinate the remaining work at the cell.
- 12 **Dev:** I will confirm stock and transport first. At the moment, I can promise a status update in ten minutes, not delivery in ten minutes.
- 13 **Inez:** That works. There is also a right-hand container on the next trolley, but it carries a hold label.
- 14 **Dev:** Leave it alone. ⁶ _____ is not a source we can use just to clear the shortage.
- 15 **Inez:** Once the correct parts arrive, we will verify the identifiers and quantities at the cell, not only the total on the trolley.
- 16 **Dev:** I will include those details on the delivery record. Will you confirm receipt so stores knows the correction reached the right location?
- 17 **Inez:** Yes. Then we can review whether the card position or the family-only list contributed to the mismatch.
- 18 **Dev:** I will bring both records. Let us fix the immediate shortage first and investigate the signal without guessing who made the mistake.
- 19 **Inez:** Good. I will tell planning we have twenty pairs now and twenty more pending, rather than eighty usable components for forty assemblies.
- 20 **Dev:** That describes the actual constraint. I will come back with the stock position and the confirmed delivery arrangement.

10

ANSWERS / EXPLANATIONS: PAGE 102

1. part-number mix | 2. shortage | 3. pick list | 4. kanban | 5. inventory balance | 6. Held stock

A family total hides a variant shortage

F Say it in a new situation

A cell needs 12 large and 12 small housings. A delivery contains 18 large and six small housings. Each assembly requires one of each.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

record / assemblies / housings / receipt

Leader: We can build only six complete ¹ _____.

Handler: We are short of six small ² _____.

Leader: Reconcile the two variants separately on the delivery ³ _____.

Handler: Verify the corrected quantities when the cell confirms ⁴ _____.

10

ANSWERS / EXPLANATIONS: PAGE 102

1. assemblies | 2. housings | 3. record | 4. receipt

Read, switch, repeat

1. Check the four answers and read the complete exchange aloud.
2. Switch roles. Read it again with the word bank covered.
3. Return to the extended conversation. Read turns 1-10, then switch roles for turns 11-20. Use the corrected expressions.

SOLO PRACTICE

Read both roles. Pause briefly at each change of speaker. Repeat any line that needed an answer check.

The first-piece report points to the wrong feature

E More workplace conversations / 1-10

Resolve a drawing-to-inspection mismatch before treating a setup as approved.

A machine setup produces a first piece for review. The report calls a 6 mm hole feature 12, but feature 12 on the current ballooned drawing is a 10 mm slot. The setup technician and inspector need to reconcile the report, drawing, and physical part. No production approval has been recorded.

PEOPLE / Nico: Setup technician | Rae: Inspector

Use each bank entry once, without changing its form. Answers follow turn 20.

WORD + PHRASE BANK

in-process checks / inspection characteristic / First-piece acceptance / datum / controlled correction / ballooned drawing

11

- 01 **Nico:** The first piece is ready, but your report marks feature twelve as failed. I measured the six-millimeter hole before bringing it over.
- 02 **Rae:** On this ¹ _____, feature twelve is the ten-millimeter slot. Let us check whether we are discussing the same feature.
- 03 **Nico:** That explains the mismatch. My setup sheet calls the hole twelve. Which drawing revision does your report identify?
- 04 **Rae:** Revision D. The ² _____ needs to match the actual slot, including its applicable requirements.
- 05 **Nico:** My sheet also says D, but the feature numbering may have been copied from an earlier report. I will bring the source file.
- 06 **Rae:** Please keep the original entry. We need a ³ _____, not a number changed to make the failure disappear.
- 07 **Nico:** Agreed. I also want to know whether the part itself meets the requirement. Fixing the label will not answer that.
- 08 **Rae:** Exactly. We will reconcile the references and inspect the correct feature using the approved method. The ⁴ _____ reference matters where the requirement calls for it.
- 09 **Nico:** Would a passing result for the hole still be useful, provided it is connected to the right characteristic?
- 10 **Rae:** Yes, if the evidence supports it. We should not discard a valid observation, but we cannot relabel it as a measurement of the slot.

The conversation, continued

E More workplace conversations / 11-20

Continue with the word bank on page 82.

- 11 **Nico:** Planning is asking whether the machine can start the full run. They see first piece complete on the schedule.
- 12 **Rae:** That should say first piece submitted. ⁵ _____ has not been recorded, and the unresolved inspection cannot be hidden by the schedule label.
- 13 **Nico:** I will correct that status now. Is this also the same as approving every part that follows?
- 14 **Rae:** No. The ⁶ _____ remain part of the production plan. A satisfactory first piece does not prove the whole run will remain satisfactory.
- 15 **Nico:** We should check the other copied feature numbers too. Otherwise this may be only the first mismatch we have noticed.
- 16 **Rae:** I agree. Let us compare the full mapping before closing the report. I will identify which results are affected and which remain correctly linked.
- 17 **Nico:** I can provide the setup sheet and the earlier template. You can show the actual inspection records beside the current drawing.
- 18 **Rae:** Good. Then the reviewer can follow the correction instead of reconstructing it from a verbal explanation later.
- 19 **Nico:** I will tell planning the setup is awaiting inspection resolution, with no production approval yet. We will give them the next status after review.
- 20 **Rae:** And we will separate the documentation mismatch from the part result. Either may need action, but they are not the same finding.

11

ANSWERS / EXPLANATIONS: PAGE 103

1. ballooned drawing | 2. inspection characteristic | 3. controlled correction | 4. datum | 5. First-piece acceptance | 6. in-process checks

A submitted first piece is shown as accepted

F Say it in a new situation

The setup team submits a first piece, but a required dimensional result is missing and the acceptance decision is still open.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

approval / accepted / run / feature

Technician: The piece is submitted, but it is not yet ¹ _____.

Inspector: Obtain the missing result for the specified ² _____.

Technician: Keep the schedule status separate from production
³ _____.

Inspector: Retain the required checks throughout the subsequent
⁴ _____.

ANSWERS / EXPLANATIONS: PAGE 103

1. accepted | 2. feature | 3. approval | 4. run

11

Read, switch, repeat

1. Check the four answers and read the complete exchange aloud.
2. Switch roles. Read it again with the word bank covered.
3. Return to the extended conversation. Read turns 1-10, then switch roles for turns 11-20. Use the corrected expressions.

SOLO PRACTICE

Read both roles. Pause briefly at each change of speaker. Repeat any line that needed an answer check.

01 Briefing and language checks

Production Flow and Daily Management

A Briefing checks / page 4

1. What was the output shortfall?

D. 160 units

Subtracting 840 completed units from the 1,000-unit target gives a 160-unit shortfall.

2. What does 84% represent here?

A. Output as a share of the stated target

The available calculation is 840 divided by 1,000, which measures output target attainment.

3. What is known about the downtime?

C. Forty minutes were logged; its cause and contribution remain under review.

The briefing gives a duration but not the cause or a complete loss attribution.

D Language checks / page 9

1. Which board entry is accurate?

D. 840 units completed; 84% of target; 160-unit shortfall.

The entry uses only the supplied output and target and keeps units distinct from time.

2. Why is OEE not established?

B. Planned time, ideal cycle time, and quality information are not supplied.

OEE combines distinct factors whose required inputs are missing from the brief.

3. A manager converts the 40-minute stop into the full 160-unit shortfall. What is missing from that attribution?

A. A supported running-rate and loss breakdown linking the stop to the output gap

The count gap is known, but minutes cannot become lost units without a relevant rate and loss basis. Other speed, quality, or operating losses may also contribute.

4. If a second shift produces 900 against a target of 1,000, what is its attainment?

A. 90%, six percentage points above 84%

900 divided by 1,000 is 90%; the difference between 90% and 84% is six percentage points.

01 Conversation answers

Production Flow and Daily Management

E Dialogue gaps

Pages 10-11

- 1. target attainment.** Turn 2. The calculation compares actual output with the target, not the separate inputs needed for OEE.
- 2. shortfall.** Turn 4. The shortfall is the target minus actual output: 1,000 minus 840 equals 160 units.
- 3. downtime.** Turn 6. Downtime is the recorded nonproducing duration, whose cause and contribution remain unconfirmed here.
- 4. Loss attribution.** Turn 8. Loss attribution assigns losses to supported contributors; the log alone does not complete that analysis.
- 5. availability.** Turn 10. Availability relates running time to planned production time, which is not supplied in this brief.
- 6. performance loss.** Turn 12. Performance loss concerns running below the defined ideal rate, requiring speed-related information beyond downtime.
- 7. Quality loss.** Turn 14. Quality loss concerns units not meeting the defined good-product criteria, information absent from the supplied count.
- 8. bottleneck.** Turn 16. A bottleneck is the system-limiting step, which must be identified from evidence rather than a local impression.
- 9. visual board.** Turn 18. The visual board should distinguish the confirmed measures, unresolved causes, and assigned next actions.
- 10. recovery plan.** Turn 20. A recovery plan is a proposal for addressing the gap, not proof that the missing output will be recovered.

F Transfer conversation

Page 12

- 1. ninety**
Lead: "Target attainment is ninety percent."
Dividing 450 by 500 gives 0.90, or ninety percent.
- 2. fifty**
Planner: "The output shortfall is fifty units."
The target of 500 minus actual output of 450 leaves fifty units.
- 3. downtime**
Lead: "Twenty minutes describes recorded downtime."
The supplied duration is nonproducing time, not a quantity of missing units.
- 4. unconfirmed**
Planner: "The cause remains unconfirmed."
The briefing provides no established explanation for the recorded stop.

02 Briefing and language checks

Lean, Waste, and Continuous Improvement

A Briefing checks / page 13

1. What was observed?

D. Eight tool-waiting episodes totaling 24 operator-minutes

The observation concerns waiting episodes and accumulated operator time, not a completed improvement or individual performance diagnosis.

2. Why is 24 not necessarily line downtime?

B. Some waiting episodes overlap, and the measure is operator-minutes.

Overlapping individual waits cannot simply be added and labeled elapsed whole-line downtime.

3. What is the proposed change?

B. A one-bench point-of-use tool trial with safety and quality checks unchanged

The scenario supplies a bounded trial proposal, not a result or permission to remove controls.

D Language checks / page 18

1. Which observation is defensible?

C. Eight tool-waiting episodes totaled 24 operator-minutes during the hour.

The statement preserves the actual event count, observation period, and accumulated time measure.

2. What is the average time per recorded episode?

C. Three operator-minutes

Dividing 24 operator-minutes by eight episodes gives three operator-minutes per episode.

3. Which proposal has suitable boundaries?

D. Trial tool placement at one bench, keep required controls, and compare waiting under comparable conditions.

The proposal defines a limited comparison without changing required controls or assuming the outcome.

4. Waiting falls at the trial bench while the shared tool is less available elsewhere. Which conclusion still needs evidence?

B. Total benefit across the connected process, including any displaced waiting

Moving delay can improve a local measure without improving the whole process. Compare relevant areas on a consistent basis before claiming overall benefit.

02 Conversation answers

Lean, Waste, and Continuous Improvement

E Dialogue gaps

Pages 19-20

1. waiting. Turn 2. Waiting describes work delayed by unavailable input, not evidence of a slow working pace.

2. operator-minute. Turn 4. An operator-minute measures one person's time; overlapping waits cannot be relabeled as elapsed whole-line downtime.

3. baseline observation. Turn 6. The baseline records the current process before a change and needs a clear time and task basis.

4. current-state map. Turn 8. A current-state map represents the existing locations, activities, and information flow before proposing a changed arrangement.

5. motion. Turn 10. Motion concerns people's movements within work, distinct from moving material between locations.

6. point-of-use storage. Turn 12. Point-of-use storage places needed items near their use, here as a proposal rather than a verified result.

7. pilot trial. Turn 14. A pilot trial is a bounded test whose conditions and retained controls must be clear before interpreting results.

8. countermeasure. Turn 16. A countermeasure is an action intended to address a problem; it does not establish achieved savings before testing.

9. value stream. Turn 18. The value stream includes connected activities, so a local change must be assessed for effects elsewhere.

10. sustainment. Turn 20. Sustainment concerns maintaining an accepted improvement through the agreed method and follow-up rather than a one-time result.

F Transfer conversation

Page 21

1. operator-minutes

Facilitator: "The accumulated waiting measure is operator-minutes."

The case adds individuals' waiting time, including overlapping episodes.

2. three

Supervisor: "Average waiting per episode is three minutes."

Twelve operator-minutes divided by four episodes gives three per episode.

3. trial

Facilitator: "The proposed tool location still needs a trial."

No evaluation of the proposed location has yet been performed.

4. savings

Supervisor: "We cannot report achieved savings yet."

A proposal has no measured improvement result before it is tested.

03 Briefing and language checks

Defects, Scrap, and Rework

A Briefing checks / page 22

1. What is the first-pass yield?

C. 90%

Nine hundred first-pass accepted units divided by 1,000 inspected units gives ninety percent.

2. How many units have approved scrap disposition?

D. 40

Only forty units are in the supplied scrap category; sixty await rework.

3. How many reworked units are already accepted?

B. None

The scenario explicitly states that no reworked units have yet been accepted.

D Language checks / page 27

1. Which report reconciles the batch correctly?

D. 900 first-pass accepted, 60 awaiting rework, and 40 scrap

The three supplied mutually exclusive states total 1,000 without counting pending recovery as accepted.

2. If all 60 later pass the required acceptance process, what would the accepted total be?

D. 960, while first-pass yield remains 90%

Later acceptance can increase final accepted quantity without changing how many passed on their first attempt.

3. One unit has three defects and is included in the sixty awaiting rework. How should the unit reconciliation treat it?

B. One pending unit, with three defects recorded in the separate defect count

The physical item appears once in its current unit category. The three defects are a different measure; neither a work plan nor multiple defects changes its acceptance or scrap status.

4. Which financial statement is supported without further data?

D. Scrap quantity and pending rework are known; their complete costs are not supplied.

The case supplies quantities but no complete costing basis for scrap or rework.

03 Conversation answers

Defects, Scrap, and Rework

03

E Dialogue gaps

Pages 28-29

- 1. scrap.** Turn 2. Scrap is the forty-unit disposition category, not all one hundred units that failed to pass initially.
- 2. mutually exclusive categories.** Turn 4. Exclusive categories place each unit in one state, allowing the total to reconcile without overlap.
- 3. first-pass yield.** Turn 6. First-pass yield is 900 divided by 1,000, excluding any later rework recovery.
- 4. rework queue.** Turn 8. The rework queue contains units awaiting the process, not units already recovered and accepted.
- 5. recovery assumption.** Turn 10. A recovery assumption describes a possible future accepted quantity and must not be reported as an achieved result.
- 6. rework.** Turn 12. Rework aims to restore conformity to requirements, whereas repair can have a different acceptance basis.
- 7. reinspection.** Turn 14. Reinspection provides the specified post-work inspection evidence and is not replaced by completion of the operation alone.
- 8. defect count.** Turn 16. Defect count can exceed defective-unit count because one unit may contain several distinct defects.
- 9. double-counting.** Turn 18. Double-counting would include recovered units in both the pending and accepted states instead of transferring their status.
- 10. material reconciliation.** Turn 20. Material reconciliation accounts for the batch across its actual states before quantities are used in cost or production summaries.

F Transfer conversation

Page 30

1. eighty-five

Analyst: "First-pass yield is eighty-five percent."

170 divided by 200 is 0.85, or eighty-five percent.

2. twenty

Manager: "The pending rework quantity is twenty."

Twenty units await rework and are not yet accepted recovered output.

3. ten

Analyst: "Approved scrap is ten units."

The scrap category contains ten units, separate from pending rework.

4. conditional

Manager: "A total of 190 accepted units remains conditional."

Reaching 190 requires all twenty pending units to pass the required acceptance process.

04 Briefing and language checks

Root Cause and Corrective Action

04

A Briefing checks / page 31

1. What changed in the relevant period?

B. The supplier and two process settings

Multiple changes occurred, so the timeline does not isolate the supplier as the cause.

2. What evidence is still unreconciled?

A. Comparable lot and process-setting records

The brief says the relevant lot and setting records have not yet been reconciled.

3. What is the status of temporary sorting?

D. An immediate screening measure, not proof of recurrence prevention

Sorting can identify affected output while leaving the underlying cause unresolved.

D Language checks / page 36

1. Which root-cause statement fits the evidence?

D. The supplier change and two setting changes remain competing explanations pending comparison.

Multiple simultaneous changes leave the cause unresolved but identify evidence that can help examine the alternatives.

2. What does sorting establish by itself?

D. A screening activity is being used; recurrence prevention is not demonstrated.

Sorting addresses detection or containment and does not by itself change the process producing the defect.

3. Which request best tests the competing supplier and setting explanations?

D. Link defect observations to actual supplier lots and both settings under comparable conditions.

Records linked to each relevant factor permit meaningful comparisons. Period labels, voting, and containment completion do not isolate a cause when several inputs changed.

4. What distinguishes an occurrence cause from a detection failure?

A. One explains production of the defect; the other explains why a check missed it.

Producing a defect and failing to detect it are distinct questions that may require different actions.

04 Conversation answers

Root Cause and Corrective Action

E Dialogue gaps

Pages 37-38

- 1. hypothesis.** Turn 2. The supplier is a possible explanation awaiting evidence, not a confirmed cause established by sequence.
- 2. correlation.** Turn 4. Correlation describes association and does not isolate causation when other relevant variables also changed.
- 3. confounding factor.** Turn 6. A confounding factor can influence the apparent relationship between the supplier switch and the defect.
- 4. change history.** Turn 8. The change history establishes which modifications occurred and when, allowing records to be compared accurately.
- 5. lot traceability.** Turn 10. Lot traceability connects specific material lots with production and defect records instead of relying on broad period labels.
- 6. containment action.** Turn 12. Containment limits impact while the cause is investigated; sorting alone does not demonstrate recurrence prevention.
- 7. occurrence cause.** Turn 14. The occurrence cause explains why the defect was produced, a question separate from whether sorting detects it.
- 8. detection failure.** Turn 16. A detection failure explains why a check missed the problem and may require an action different from the production cause.
- 9. effectiveness check.** Turn 18. An effectiveness check evaluates whether the action achieved the intended prevention result rather than merely being documented.
- 10. closure criteria.** Turn 20. Closure criteria specify the evidence required before the issue can be accepted as resolved.

F Transfer conversation

Page 39

1. hypothesis

Engineer: "The material explanation is still a hypothesis."

Multiple changes occurred, so the material has not been established as the cause.

2. factor

Reviewer: "The setting change is a potential confounding factor."

The other changing variable can affect the apparent material-defect relationship.

3. action

Engineer: "Temporary sorting is a containment action."

Screening limits impact while the cause remains under investigation.

4. evidence

Reviewer: "Recurrence prevention still needs effectiveness evidence."

Active containment does not establish that a corrective action prevents the defect from recurring.

05 Briefing and language checks

Maintenance, Reliability, and Changeover

A Briefing checks / page 40

1. When do the two demands overlap?

B. From 13:00 to 14:00

Maintenance starts at 13:00 while the current rush-order schedule needs the line until 14:00.

2. What is known about maintenance deferral?

D. No supporting technical assessment or authorization is supplied.

The brief leaves the assessment and decision open rather than approving a schedule change.

3. What is the alternative-line status?

D. Possible but unconfirmed in suitability

A potential option does not establish the line's technical suitability, readiness, or allocation.

D Language checks / page 45

1. Which schedule statement is accurate?

A. There is a one-hour overlap, but the maintenance scope remains two hours unless properly reassessed.

The overlap calculation does not shorten the planned work or authorize a schedule change.

2. Which alternative-line statement is best?

B. Assess suitability, setup, readiness, and allocation before committing the order.

A credible contingency needs more than the possibility of unused capacity.

3. Which reply preserves authority?

B. We can present deferral for assessment, but it is not authorized in the current information.

The supplied case contains no technical basis or authorized deferral decision.

4. The maintenance operation is marked finished. What remains a separate status to confirm?

D. Required checks and authorized return to service

Finishing the work does not automatically establish that the required checks and return-to-service decision are complete. The original schedule facts do not replace those controls.

05 Conversation answers

Maintenance, Reliability, and Changeover

E Dialogue gaps

Pages 46-47

1. maintenance window. Turn 2. The maintenance window is the reserved period from 13:00 to 15:00, which conflicts with the order schedule.

2. deferral. Turn 4. Deferral is a proposed postponement requiring the relevant assessment and authorization, neither of which is supplied here.

3. reliability. Turn 6. Reliability concerns performance over defined conditions and time, not a guarantee inferred from one recent operating period.

4. work order. Turn 8. The work order identifies the defined maintenance task and its controlled scope for the scheduling assessment.

5. capacity contingency. Turn 10. A capacity contingency is a conditional alternative whose suitability and readiness still need confirmation.

6. changeover. Turn 12. A changeover transitions equipment to another job, while the original task concerns maintenance.

7. External setup. Turn 14. External setup consists of eligible preparation while equipment runs; not every task can safely or appropriately be moved there.

8. spare-part readiness. Turn 16. Spare-part readiness establishes whether the correct needed parts are available for the planned scope.

9. schedule dependency. Turn 18. A schedule dependency identifies a condition that must be resolved before the corresponding plan can be relied on.

10. Return to service. Turn 20. Return to service is the authorized restoration of operational use after the relevant work and checks, not merely task completion.

F Transfer conversation

Page 48

1. one

Planner: "The schedules overlap by one hour."

The shared period is 10:00 to 11:00, a one-hour overlap.

2. deferral

Technician: "Moving maintenance would be a proposed deferral."

Postponement is not approved merely because an order conflicts with the window.

3. contingency

Planner: "The second line is a capacity contingency."

The alternative remains conditional until its suitability and readiness are confirmed.

4. suitability

Technician: "Before using it, confirm its suitability."

The briefing specifically leaves the second line's capability for the job unconfirmed.

05

06 Briefing and language checks

EHS and Safety Communication

A Briefing checks / page 49

1. Where is this exchange taking place?

A. In a classroom safety exercise with no equipment work underway

The brief explicitly describes a classroom exercise and supplies no live-work authorization.

2. Which clarification is needed before this classroom card could support the intended technical discussion?

B. Identified equipment, energy sources, and the current applicable approved procedure

The card lacks the equipment-specific basis needed to interpret it. A signature or general statement does not supply that basis, and the exercise authorizes no actual work.

3. What must not be invented?

A. An isolation sequence for an unidentified machine

The task is to stop and clarify communication, not create a machine procedure without the required basis.

D Language checks / page 54

1. Which response best fits the ambiguous card?

B. Pause and obtain the current applicable equipment-specific procedure through the authorized route.

The missing context requires clarification by the proper process, not invented operational instructions.

2. What does switching off establish by itself?

C. Normal operation may be stopped; complete energy isolation is not established.

Shutdown and complete hazardous-energy control are distinct, so the broader conclusion is unsupported.

3. What can read-back confirm?

B. Shared understanding of the clarified message, not the physical isolated state

Repeating information checks communication and does not replace technical verification or role requirements.

4. Which closing statement is accurate?

D. The ambiguous classroom card is referred for correction; no equipment work or restart is authorized.

The exercise identifies a communication problem without supplying live-work or restart authorization.

06 Conversation answers

EHS and Safety Communication

E Dialogue gaps

Pages 55-56

1. stop-work concern. Turn 2. The concern calls for a pause at the ambiguous instruction rather than guessing a work sequence.

2. Shutdown. Turn 4. Shutdown stops normal operation but does not establish that every hazardous energy source has been controlled.

3. Stored energy. Turn 6. Stored energy can remain after normal supply or operation stops, so it cannot be dismissed from a generic off instruction.

4. equipment-specific procedure. Turn 8. The equipment-specific procedure supplies the approved instructions for identified equipment rather than a generic classroom shortcut.

5. authorized employee. Turn 10. The authorized employee has the assigned role and required training for the relevant energy-control work.

6. Tagout. Turn 12. Tagout uses warning tags within the applicable system and must not be assumed to provide the same physical restraint as a lock.

7. read-back. Turn 14. Read-back checks shared understanding of critical information but does not establish a physical equipment condition.

8. Verification of isolation. Turn 16. Verification of isolation requires the approved technical method, which has not occurred in this classroom discussion.

9. EHS. Turn 18. EHS identifies the environment, health, and safety function involved in addressing the safety-related training issue.

10. restart authorization. Turn 20. Restart authorization is a separate required permission and is not created by completing a classroom clarification.

F Transfer conversation

Page 57

1. missing

Learner: "The equipment identity is missing."

The card omits the specific machine, so its identity cannot be assumed.

2. procedure

Trainer: "We must obtain the applicable approved procedure."

A generic card cannot replace the required equipment-specific instructions.

3. exercise

Learner: "This discussion remains a classroom exercise."

The case supplies no actual equipment work or servicing authorization.

4. restart

Trainer: "It does not authorize a restart."

A training clarification does not provide permission to resume equipment operation.

07 Briefing and language checks

Supplier Quality and Incoming Materials

A Briefing checks / page 58

1. What is the revision C range?

D. 11.90 to 12.10 mm

A nominal 12.00 mm with a tolerance of 0.10 gives limits of 11.90 and 12.10.

2. How does 12.18 mm compare with revision B's stated range?

A. It lies within 12.10 to 12.30 mm.

The measurement is within the stated B limits, but that comparison does not establish the governing revision.

3. What remains unconfirmed?

A. Which drawing revision governs this order

The brief supplies both references but not the agreed revision applicable to the order.

D Language checks / page 63

1. What is the difference between 12.18 mm and revision C's upper limit?

B. 0.08 mm above the limit

The upper limit is 12.10 mm, and 12.18 minus 12.10 equals 0.08 mm.

2. Which update keeps the numerical result separate from the unresolved governing requirement?

B. 12.18 meets B's 12.10-12.30 range and exceeds C's upper limit by 0.08; applicability and release remain open.

C's upper limit is 12.10, making the excess 0.08, not the distance from nominal. Neither numerical fit nor revision chronology establishes applicability or release authority.

3. Which evidence best addresses the mismatch?

B. Order references, acknowledgment, drawing versions, and change effectivity records

The relevant records connect the agreed requirement and change timing to the particular order and lot.

4. How should the report be corrected?

C. Add the missing revision and resolution through the controlled record process while retaining traceability.

The correction must preserve the measured result and traceable requirement basis rather than obscure the discrepancy.

07 Conversation answers

Supplier Quality and Incoming Materials

E Dialogue gaps

Pages 64-65

- 1. drawing revision.** Turn 2. The drawing revision identifies the requirement version needed to interpret the inspection result.
- 2. lower specification limit.** Turn 4. Subtracting 0.10 from the B nominal 12.20 gives the lower permitted value of 12.10 mm.
- 3. upper specification limit.** Turn 6. Adding 0.10 to the C nominal 12.00 gives the upper permitted value of 12.10 mm.
- 4. measured value.** Turn 8. The measured value remains 12.18 mm; differing reference limits change its interpretation, not the recorded number.
- 5. order acknowledgment.** Turn 10. The acknowledgment records the supplier's received or accepted terms and can be compared with the purchase order references.
- 6. effectivity.** Turn 12. Effectivity specifies when or to which population the technical change applies, resolving more than simple revision chronology.
- 7. supplier flow-down.** Turn 14. Supplier flow-down concerns how applicable requirements were communicated through the supply chain.
- 8. acceptance basis.** Turn 16. The acceptance basis remains unresolved while the applicable revision is unknown, so a favorable B comparison cannot settle release.
- 9. deviation request.** Turn 18. A deviation request seeks an authorized departure and is not itself approval of that departure.
- 10. release decision.** Turn 20. The release decision is the authorized result of the review, not an assumption drawn from one favorable comparison.

F Transfer conversation

Page 66

1. A

Engineer: "The result is within revision A limits."

5.08 lies between the supplied A limits of 4.90 and 5.10.

2. limit

Supplier: "It exceeds revision B's upper limit."

5.08 is above the B upper permitted value of 5.05.

3. unconfirmed

Engineer: "The applicable revision remains unconfirmed."

The case does not establish which reference governs the order.

4. released

Supplier: "The lot has not been released."

The material remains held, so neither comparison supplies release authorization.

08 Briefing and language checks

Shift Handoffs and Escalation

A Briefing checks / page 67

1. Which checks remain open?

A. Check 7 label-count reconciliation and check 8 final quality review

The brief identifies two specific required checks and their dependency.

2. What is required before lot release?

B. Completion of all eight required checks under the site procedure

The supplied site procedure requires every release check, not a partial completion threshold.

3. What has Elena not yet confirmed at the start?

D. Receipt of the handoff

A named recipient is not proof that the information has been received and understood.

D Language checks / page 72

1. Which handoff line is most useful?

A. L27: checks 1-6 complete; 7 reconciliation and 8 final review open; lot not released.

The line identifies the actual completed and open work and preserves the stated release restriction.

2. Elena confirms receipt and explicitly accepts follow-up responsibility. What may the record state?

A. Handoff received and ownership accepted; checks 7 and 8 remain open and the lot is not released.

Receipt and explicit acceptance establish communication and ownership. They do not complete the checks, remove their dependency, or authorize the lot's release.

3. The teams agree an update at 09:00. Which interpretation is safest and most precise?

A. A status report is due at 09:00; completion is not guaranteed unless separately established.

An agreed checkpoint concerns reporting status and should not be turned into an unsupported completion promise.

4. Which task order matches the brief?

A. Complete label reconciliation before the dependent final quality review.

The brief explicitly makes final quality review dependent on the completed reconciliation.

08 Conversation answers

Shift Handoffs and Escalation

E Dialogue gaps

Pages 73-74

- 1. open item.** Turn 2. An open item is unfinished work that must be named, rather than hidden behind a readiness percentage.
- 2. dependency.** Turn 4. The dependency requires label reconciliation to be completed before the final quality review can be completed.
- 3. attachment.** Turn 6. Attachments are the supporting records whose availability must be confirmed, not assumed from a sent message.
- 4. release checklist.** Turn 8. The release checklist defines the required checks, all of which remain necessary before the stated release decision.
- 5. status qualifier.** Turn 10. A status qualifier such as not released states the limits of the current condition rather than offering vague reassurance.
- 6. read-back.** Turn 12. Read-back repeats the critical lot status, open checks, and restriction to confirm shared understanding.
- 7. handover acceptance.** Turn 14. Handover acceptance confirms responsibility for follow-up and does not establish technical completion of the remaining checks.
- 8. next checkpoint.** Turn 16. The next checkpoint is an agreed status-review time, not a guaranteed completion or release time.
- 9. escalation trigger.** Turn 18. The trigger identifies the missing evidence condition that requires referral through the site's responsible process.
- 10. audit trail.** Turn 20. The audit trail records who knew and accepted what, alongside later evidence and decisions, without rewriting earlier uncertainty.

F Transfer conversation

Page 75

1. open

Outgoing lead: "The final review remains open."

Three completed checks leave the fourth review unfinished.

2. responsibility

Noor: "I accept follow-up responsibility."

Noor accepts ownership of the follow-up rather than declaring the review complete.

3. time

Outgoing lead: "Sixteen hundred is the agreed update time."

The checkpoint is a reporting time, not a guaranteed completion time.

4. release

Noor: "The handoff does not itself authorize release."

An accepted handoff cannot replace the remaining required review or release decision.

09 Conversation answers

The gauge passes today, but what about yesterday?

E Extended conversation

Pages 76-77

1. as-found results. Turn 2. As-found results describe the instrument before adjustment, which is the relevant starting point for assessing earlier measurements.

2. as-left results. Turn 4. As-left results describe the instrument after the calibration work or adjustment and do not establish its earlier condition.

3. asset identifier. Turn 6. The asset identifier connects the specific instrument to its inspection records, avoiding assumptions based on a tool description alone.

4. measurement uncertainty. Turn 10. Measurement uncertainty describes the dispersion associated with a measured value; it contributes to evaluating whether the measurement is fit for its intended decision.

5. reinspection. Turn 12. Reinspection is a new examination under an established plan; its results must remain distinguishable from the original inspection.

6. fitness for purpose. Turn 16. Fitness for purpose means the measurement capability is adequate for the intended task, not merely that a calibration label exists.

F Transfer exchange

Page 78

1. measurement

Supervisor: The label does not establish fitness for this measurement.

A calibration label alone does not show suitability for the particular measurement task.

2. limits

Engineer: Compare the instrument capability with the feature and its limits.

The relevant feature limits help determine the measurement capability required for the decision.

3. inspection

Supervisor: Keep earlier results separate from any new inspection.

New inspection results must not silently overwrite the original evidence.

4. instrument

Engineer: Record the basis for selecting the replacement instrument.

The selection should identify the actual instrument and why it is suitable.

09

Background reading: [NIST: Metrological Traceability Policy](#). The conversation is an original fictional teaching case.

10 Conversation answers

The right total, the wrong replenishment mix

E Extended conversation

Pages 79-80

- 1. part-number mix.** Turn 2. The part-number mix describes the quantities of different variants; the correct combined total does not establish the required mix.
- 2. shortage.** Turn 4. The shortage is twenty right-hand brackets relative to the forty required; excess left-hand brackets do not substitute for them.
- 3. pick list.** Turn 6. The pick list records the items and quantities requested for retrieval, helping locate where the variant mismatch entered the process.
- 4. kanban.** Turn 8. A kanban is a signal authorizing production or movement within a defined pull system; it must identify the relevant item and quantity.
- 5. inventory balance.** Turn 10. The inventory balance must reflect the actual quantity and location after the physical movement, rather than retaining an incorrect delivery total.
- 6. Held stock.** Turn 14. Held stock is restricted pending the applicable decision; physical availability does not mean it is available for production.

F Transfer exchange

Page 81

1. assemblies

Leader: We can build only six complete assemblies.

Six small housings limit production to six assemblies despite the correct overall count.

2. housings

Handler: We are short of six small housings.

The requirement is twelve small housings but only six have arrived.

3. record

Leader: Reconcile the two variants separately on the delivery record.

A combined total hides the incorrect mix and must not replace variant-level quantities.

4. receipt

Handler: Verify the corrected quantities when the cell confirms receipt.

Receipt confirmation establishes that the corrected delivery reached the intended cell.

10

Background reading: [Lean Enterprise Institute: Kanban](#). The conversation is an original fictional teaching case.

11 Conversation answers

The first-piece report points to the wrong feature

E Extended conversation

Pages 82-83

1. ballooned drawing. Turn 2. A ballooned drawing numbers the features to connect them with inspection results; the number must refer to the correct feature and revision.

2. inspection characteristic. Turn 4. The inspection characteristic is the specific feature or requirement being evaluated, not merely a number copied between documents.

3. controlled correction. Turn 6. A controlled correction preserves the original record and documents the authorized change, rather than silently replacing an inconsistent result.

4. datum. Turn 8. A datum is the theoretically exact reference used for applicable dimensional or geometric requirements; the reference must match the drawing and method.

5. First-piece acceptance. Turn 12. First-piece acceptance is the recorded determination that the required setup checks meet their criteria; making or submitting the piece does not establish acceptance.

6. in-process checks. Turn 14. In-process checks monitor the subsequent production according to the established plan; first-piece acceptance does not remove those requirements.

F Transfer exchange

Page 84

1. accepted

Technician: The piece is submitted, but it is not yet accepted.

Submission is a handover milestone and does not establish the inspection decision.

2. feature

Inspector: Obtain the missing result for the specified feature.

The required feature result is still needed before the inspection can be resolved.

3. approval

Technician: Keep the schedule status separate from production approval.

A schedule label cannot substitute for the authorized production decision.

4. run

Inspector: Retain the required checks throughout the subsequent run.

Acceptance of a first piece does not remove the established in-process inspection requirements.

Find the move you need

Use these as precise language models. Replace details only when you know the actual facts, authority, and agreed conditions.

01 Production Flow and Daily Management

Lead with the result: We completed 840 units against a target of 1,000.

Correct the label: The supplied figures do not support an OEE calculation.

Ask for the time basis: What planned production time was used in this calculation?

Full phrase set: pages 7-8 | Dialogue: pages 10-11

02 Lean, Waste, and Continuous Improvement

Describe the delay: Operators waited for the shared tool during eight observed episodes.

Challenge the attribution: The observation does not establish that operator speed caused the delay.

Bound the claim: No improvement result is available yet.

Full phrase set: pages 16-17 | Dialogue: pages 19-20

03 Defects, Scrap, and Rework

Define the population: The report covers 1,000 inspected units from this batch.

State the scrap decision: Forty units have approved scrap disposition.

Report the current state: No reworked units have been accepted yet.

Full phrase set: pages 25-26 | Dialogue: pages 28-29

04 Root Cause and Corrective Action

State the timing: The defect first appears after the supplier switch in the available records.

Request comparable records: Link the defect data to the actual supplier lots and settings.

Keep the evidence organized: Stratify the observations by the relevant conditions.

Full phrase set: pages 34-35 | Dialogue: pages 37-38

Keep the conversation productive

Use these as precise language models. Replace details only when you know the actual facts, authority, and agreed conditions.

05 Maintenance, Reliability, and Changeover

State the overlap: The order needs the line until 14:00, while maintenance starts at 13:00.

Respect decision ownership: The responsible decision maker needs the technical assessment before approving a change.

Preserve required scope: Do not remove required work just to fit the schedule.

Full phrase set: pages 43-44 | Dialogue: pages 46-47

06 EHS and Safety Communication

State the concern: Stop here: the instruction does not identify the equipment or isolation procedure.

Identify missing context: Which equipment and approved procedure does this instruction refer to?

Keep training separate: This classroom exercise does not authorize actual servicing.

Full phrase set: pages 52-53 | Dialogue: pages 55-56

07 Supplier Quality and Incoming Materials

State the result fully: The reported dimension is 12.18 millimeters.

Read the C limits: Revision C permits 11.90 to 12.10 millimeters.

Trace the agreement: Compare the purchase order with the supplier acknowledgment.

Full phrase set: pages 61-62 | Dialogue: pages 64-65

08 Shift Handoffs and Escalation

Identify the lot: This handoff concerns lot L27.

Describe the dependency: The final review depends on completion of the reconciliation.

Confirm receipt: Please confirm you can access the attached records.

Full phrase set: pages 70-71 | Dialogue: pages 73-74

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

5S sustain 5S practices	15
8D prepare an 8D report	33
acceptance basis state the acceptance basis	60
acceptance status verify acceptance status	23
acknowledgment obtain acknowledgment	69
actual output report actual output	5
affected employee inform affected employees	50
applicable revision establish the applicable revision	60
attachment verify the attachment	68
audit trail preserve the audit trail	69
authorized employee identify the authorized employee	50
availability assess equipment availability	6
backlog prioritize the maintenance backlog	41
baseline observation record a baseline observation	15
bottleneck identify the bottleneck	5
capacity contingency assess a capacity contingency	42
cause-and-effect diagram build a cause-and-effect diagram	32
change history reconstruct the change history	32
changeover plan the changeover	42
closed-loop communication use closed-loop communication	69
closure criteria agree closure criteria	33
comparative evidence obtain comparative evidence	32
completed check verify completed checks	68
condition monitoring review condition-monitoring data	41

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

confounding factor identify confounding factors	32
constraint manage the constraint	5
containment action implement a containment action	33
continuity maintain continuity	69
controlled comparison authorize a controlled comparison	33
controlled copy obtain the controlled copy	60
corrective action implement corrective action	33
corrective maintenance plan corrective maintenance	41
correlation distinguish correlation from causation	32
cost of poor quality assess cost of poor quality	24
countermeasure test a countermeasure	15
critical asset identify critical assets	41
current-state map create a current-state map	14
cycle time measure cycle time	5
daily management support daily management	6
defect classify a defect	23
defect count distinguish defect count	24
defective unit count defective units	23
deferral assess a maintenance deferral	42
dependency identify a task dependency	68
detection failure investigate a detection failure	33
deviation request review a deviation request	60
disposition approve a disposition	23
document discrepancy resolve a document discrepancy	60

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

double-counting	prevent double-counting	24
downtime	log downtime	6
drawing revision	confirm the drawing revision	59
due time	distinguish due time from update time	69
effectiveness check	define an effectiveness check	33
effectivity	confirm change effectivity	60
EHS	raise an EHS concern	50
energy isolation	follow the energy-isolation procedure	50
engineering change notice	trace the engineering change notice	60
engineering drawing	consult the engineering drawing	59
equipment-specific procedure	consult the equipment-specific procedure	51
escalation trigger	define an escalation trigger	69
escape point	identify the escape point	33
excess inventory	reduce excess inventory	14
exclusion zone	respect the exclusion zone	51
external setup	separate external setup tasks	42
failure mode	identify a failure mode	42
final quality review	complete the final quality review	68
final yield	state the final-yield basis	23
first-pass yield	calculate first-pass yield	23
five whys	use five whys carefully	32
future-state map	develop a future-state map	15
gemba	observe at the gemba	14
handoff record	update the handoff record	68

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

handover acceptance	record handover acceptance	69
hazard	identify a hazard	50
hazardous energy	recognize hazardous energy	50
hierarchy of controls	apply the hierarchy of controls	51
hold status	maintain hold status	24
hypothesis	test a hypothesis	32
incoming shift	inform the incoming shift	68
inspection characteristic	identify the inspection characteristic	59
inspection report	correct the inspection report	59
internal setup	identify internal setup tasks	42
job hazard analysis	complete a job hazard analysis	51
kaizen	run a kaizen event	14
kanban	use a kanban signal	15
label reconciliation	complete label reconciliation	68
lead time	reduce lead time	5
lean manufacturing	apply lean manufacturing	14
line balance	improve line balance	5
lockout	follow the lockout procedure	50
loss attribution	verify loss attribution	6
lot identity	preserve lot identity	60
lot traceability	maintain lot traceability	32
lower specification limit	check the lower specification limit	59
maintainability	improve maintainability	41
maintenance window	protect the maintenance window	41

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

material reconciliation	perform material reconciliation	24
mean time between failures	interpret mean time between failures	41
mean time to repair	state the mean-time-to-repair basis	41
measured value	record the measured value	59
measurement unit	state the measurement unit	59
motion	analyze unnecessary motion	14
mutually exclusive categories	use mutually exclusive categories	24
named owner	assign a named owner	69
near miss	report a near miss	51
next checkpoint	set the next checkpoint	69
nominal dimension	identify the nominal dimension	59
nonconforming unit	identify nonconforming units	23
occurrence cause	verify the occurrence cause	33
open item	identify open items	68
operator-minute	report operator-minutes	15
order acknowledgment	compare the order acknowledgment	59
outgoing shift	brief the outgoing shift	68
overall equipment effectiveness	calculate overall equipment effectiveness	6
overprocessing	identify overprocessing	14
overproduction	prevent overproduction	14
pending decision	flag a pending decision	69
performance loss	analyze performance losses	6
permit to work	verify permit-to-work requirements	51
personal protective equipment	select required personal protective equipment	51

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

pilot trial define a pilot trial	15
planned stop distinguish planned stops	6
point-of-use storage trial point-of-use storage	15
predictive maintenance use predictive maintenance	41
preventive maintenance schedule preventive maintenance	41
process parameter review process parameters	32
production target set a production target	5
pull system design a pull system	15
purchase order review the purchase order	59
quality escape investigate a quality escape	24
quality loss quantify quality losses	6
read-back request a read-back	51
read-back request a read-back	69
receiving record trace the receiving record	60
record reference provide the record reference	68
recovery assumption qualify the recovery assumption	24
recovery plan review the recovery plan	6
recurrence monitor recurrence	33
reinspection complete reinspection	24
release checklist review the release checklist	68
release decision document the release decision	60
reliability assess equipment reliability	41
repair distinguish repair from rework	23
residual energy assess residual energy	50

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

restart authorization confirm restart authorization	51
return to service confirm return to service	42
rework perform authorized rework	23
rework labor record rework labor	24
rework queue monitor the rework queue	23
root cause establish the root cause	32
safety data sheet consult the safety data sheet	51
schedule dependency identify schedule dependencies	42
scrap record approved scrap	23
scrap cost calculate scrap cost	24
scrap rate report the scrap rate	23
setup time measure setup time	42
shift handoff conduct a shift handoff	68
shortfall quantify the shortfall	5
shutdown distinguish shutdown from isolation	50
SMED apply SMED principles	42
sorting document temporary sorting	33
spare-part readiness confirm spare-part readiness	42
standard work update standard work	15
status qualifier preserve status qualifiers	69
stop-work concern raise a stop-work concern	51
stored energy address stored energy	50
stratification stratify by supplier and setting	32
superseded revision identify a superseded revision	60

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

supplier flow-down	verify supplier flow-down	60
sustainment	plan sustainment	15
symptom	distinguish a symptom from a cause	32
tagout	understand tagout limitations	50
takt time	calculate takt time	5
target attainment	calculate target attainment	5
throughput	measure throughput	5
tier meeting	escalate through a tier meeting	6
tolerance	apply the stated tolerance	59
transportation	reduce unnecessary transportation	14
unexpected startup	prevent unexpected startup	50
unit count	verify the unit count	24
unplanned stop	investigate unplanned stops	6
upper specification limit	check the upper specification limit	59
value stream	map the value stream	14
verification evidence	retain verification evidence	33
verification of isolation	require verification of isolation	51
visual board	update the visual board	6
waiting	reduce waiting time	14
waste	identify process waste	14
wear indicator	monitor wear indicators	42
work in process	control work in process	5
work order	review the work order	41
work-cell layout	assess the work-cell layout	15

Original cases. Clear boundaries.

These are original fictional teaching conversations, not transcripts of real workers. Names, figures, organizations, and incidents are invented. Sources support terminology and communication aims, not the fictional results or an endorsement of this book.

[1] National Institute of Standards and Technology. Value Stream Mapping.

[Open the original source](#)

Background on mapping material and information flow to identify improvement opportunities. The observations, pilot proposals, and datasets here are original. Checked 10 October 2026.

[2] Lean Enterprise Institute. Single Minute Exchange of Die.

[Open the original source](#)

Background on changeover reduction and the distinction between internal and external setup. No example authorizes a machine operation. Checked 10 October 2026.

[3] American Society for Quality. Quality Glossary.

[Open the original source](#)

Terminology reference. Definitions and practice cases in this book are independently written for language learning, not certification or a quality-system standard. Checked 10 October 2026.

[4] US Occupational Safety and Health Administration. Control of Hazardous Energy.

[Open the original source](#)

Background on the need for an appropriate energy-control program, procedures, and training. The book deliberately supplies no machine-specific isolation sequence. Checked 10 October 2026.

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